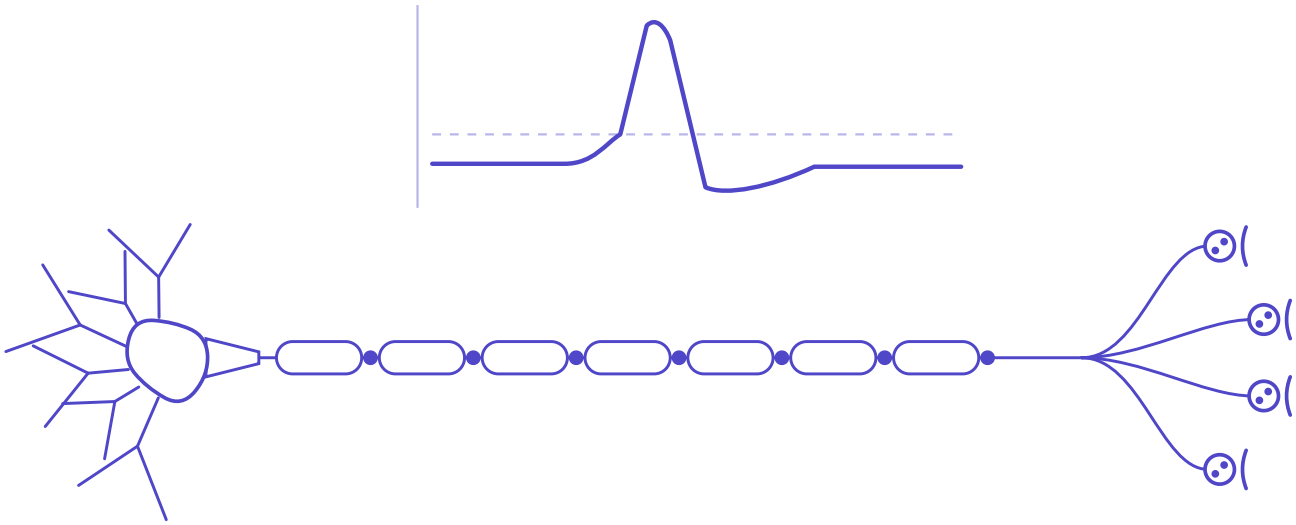


3 TIMED TESTS · 50 QUESTIONS EACH

Neurology

Practice tests of USMLE-style questions drawn from the website quizzes, with answers and explanations.

For educational use only. This is a study aid, not a clinical reference.

Instructions

60 minutes
for each test

50 questions
in each test

About 72 seconds
for each question

Every question comes from the website. These tests are built from the quizzes on the Clinical Conditions website, 150 of its 522 questions, so if you have done the website quizzes you will meet questions you have seen before. For the truest measure, take a test before using the website quiz for these topics, and treat a remembered answer as revision rather than a sign of mastery.

TAKING A TEST

- Take each test in one sitting, with a timer set to 60 minutes. Do not pause the timer.
- Each question has five options and one best answer. Choose the single best option, even if more than one seems reasonable.
- Fill in the answer sheet after the last question of each test, or use a separate piece of paper.
- Do not use notes, the website, or other references. The abbreviation list on the next page is allowed.
- There is no penalty for guessing, so answer every question. An unanswered question counts as incorrect.
- When time ends, stop, even if you have not finished.

ABOUT THE TESTS

- No question appears in more than one test, so the three tests can be taken in any order.
- Every test has 15 physiology and 35 clinical questions, covering all 24 physiology topics and all 29 clinical topics, with a similar mix of Step 1 questions (mechanisms, microbiology, and pathology) and Step 2 questions (diagnosis, next step, and management).
- Questions are in random order, and the topic being tested is not shown until the answers.

SCORING AND REVIEW

- Your score is the number of correct answers. The answer key at the start of each answer section has the same layout as the answer sheet, so you can mark one against the other.
- Then read the explanations. Each one names the topic and step, so you can see what to revisit.
- Answers follow the course material. Where a guideline differs, the explanation says so under "Additional information".

CONTENTS

Test A Questions	Test B Questions	Test C Questions
Test A Answer sheet	Test B Answer sheet	Test C Answer sheet
Test A Answers	Test B Answers	Test C Answers

Abbreviations

ALLOWED DURING A TEST

ACA Anterior cerebral artery

AComm Anterior communicating artery

ADC Apparent diffusion coefficient

AFP Alpha-fetoprotein

AICA Anterior inferior cerebellar artery

ALS Amyotrophic lateral sclerosis

ANA Antinuclear antibody

ANP Atrial natriuretic peptide

APOE Apolipoprotein E

APOE4 Apolipoprotein E ε4 allele

APP Amyloid precursor protein

AQP4 Aquaporin-4

AQP4-IgG Aquaporin-4 immunoglobulin G antibody

ARIA Amyloid-related imaging abnormalities; Amyloid-related imaging abnormalities: edema or microhemorrhage

Aβ42 Amyloid beta 42, the 42-amino-acid form

B12 Vitamin B12 (cobalamin)

BBB Blood-brain barrier

BP Blood pressure

BPPV Benign paroxysmal positional vertigo

bvFTD Behavioral variant frontotemporal dementia

C5 Complement component 5

C9orf72 Chromosome 9 open reading frame 72 gene

CBD Corticobasal degeneration

CK Creatine kinase

CK Creatine kinase

CMV Cytomegalovirus

CN Cranial nerve

CN III Cranial nerve III (oculomotor); Cranial nerve III (the oculomotor nerve)

CN VI Cranial nerve VI (abducens)

CN VII Cranial nerve VII (facial)

CN X Cranial nerve X (the vagus nerve)

COMT Catechol-O-methyltransferase

COPD Chronic obstructive pulmonary disease

CRAO Central retinal artery occlusion

CRVO Central retinal vein occlusion

CSF Cerebrospinal fluid

CT Computed tomography

CXR Chest x-ray

CYP3A4 Cytochrome P450 3A4

D1 Dopamine receptor 1

D2 Dopamine receptor 2

D3 Dopamine receptor 3

DAT Dopamine transporter

DBS Deep brain stimulation

DDC DOPA decarboxylase

DLB Dementia with Lewy bodies

DWI Diffusion-weighted imaging

ECG Electrocardiogram

EDNRB Endothelin receptor type B gene

EEG Electroencephalogram

ESR Erythrocyte sedimentation rate

FLAIR Fluid-attenuated inversion recovery

GABA Gamma-aminobutyric acid

GABA-A GABA type A receptor, a chloride channel

GCS Glasgow Coma Scale

GJB2 Gap junction beta 2 gene (connexin 26)

GPe Globus pallidus externus

GPi Globus pallidus internus

HINTS Head impulse, nystagmus, and test of skew

HLA-B*1502 Human leukocyte antigen B*15:02 allele; A human leukocyte antigen allele that predicts carbamazepine SJS

HMG-CoA 3-hydroxy-3-methylglutaryl coenzyme A

HR Heart rate

HTT Huntingtin gene

ICA Internal carotid artery

ICH Intracerebral hemorrhage

ICP Intracranial pressure

INO Internuclear ophthalmoplegia

IOP Intraocular pressure

IV Intravenous

IVIG Intravenous immunoglobulin

M1 Muscarinic acetylcholine receptor 1; Primary motor cortex

MAO-B Monoamine oxidase B

MCA Middle cerebral artery

MITF Melanocyte-inducing transcription factor gene

MLF Medial longitudinal fasciculus

MOG-IgG Myelin oligodendrocyte glycoprotein immunoglobulin G antibody

MOGAD MOG antibody-associated disease

MRA Magnetic resonance angiography

MRI Magnetic resonance imaging

MS Multiple sclerosis

MSA Multiple system atrophy

NMDA N-methyl-D-aspartate; N-methyl-D-aspartate glutamate receptor

NMOSD Neuromyelitis optica spectrum disorder

NPH Normal pressure hydrocephalus

NSAIDs Nonsteroidal anti-inflammatory drugs

PaCO2 Arterial partial pressure of carbon dioxide

PAX3 Paired box 3 gene

PComm Posterior communicating artery

PICA Posterior inferior cerebellar artery

PSP Progressive supranuclear palsy

PSP-bvFTD Progressive supranuclear palsy, behavioral variant frontotemporal dementia

PSP-CBS Progressive supranuclear palsy, corticobasal syndrome

PSP-P Progressive supranuclear palsy, parkinsonism predominant

PSP-PNFA Progressive supranuclear palsy, progressive nonfluent aphasia

PSP-RS Progressive supranuclear palsy, Richardson syndrome

QT The interval from the start of the QRS complex to the end of the T wave

QT The QT interval on the ECG

RAAS Renin-angiotensin-aldosterone system

RAPD Relative afferent pupillary defect

REM Rapid eye movement

RTX Rituximab

SAH Subarachnoid hemorrhage

SNc Substantia nigra pars compacta

SNHL Sensorineural hearing loss

SPECT Single-photon emission computed tomography

STN Subthalamic nucleus

SV2A Synaptic vesicle protein 2A; Synaptic vesicle glycoprotein 2A

SWI Susceptibility-weighted imaging

TPMT Thiopurine methyltransferase

TPMT Thiopurine methyltransferase

V1 Ophthalmic division of the trigeminal nerve; Primary visual cortex

VA Ventral anterior nucleus of the thalamus

VEGF Vascular endothelial growth factor

VL Ventral lateral nucleus of the thalamus

VMAT2 Vesicular monoamine transporter 2

Test A

50 questions | 60 minutes | Start the timer

1 A 28-year-old woman has 3 days of blurred vision in her left eye and pain when she moves it. Colors look washed out, and she has a left RAPD. The optic disc looks normal.

Which investigation is most important next?

- (A) Electroretinogram
- (B) Fluorescein angiography of the retina and choroid
- (C) Gonioscopy
- (D) MRI of the brain and orbits with gadolinium
- (E) Temporal artery biopsy

2 A head CT shows a crescent-shaped collection along the left convexity that crosses the coronal suture but stops at the falx.

What is this collection?

- (A) Cerebral abscess
- (B) Subdural hematoma
- (C) Subarachnoid hemorrhage
- (D) Epidural hematoma
- (E) Intracerebral hemorrhage

3 A 27-year-old man has painful vision loss in both eyes over a week. Both optic discs are swollen, and MRI shows enhancement of most of each optic nerve, with perineural sheath enhancement.

Which antibody test is most likely to be positive?

- (A) Anti-acetylcholine receptor
- (B) MOG-IgG
- (C) Anti-NMDA receptor
- (D) AQP4-IgG
- (E) ANA

4 A student compares the lateral and medial spinothalamic pathways.

What does the medial pathway mainly convey?

- (A) Unconscious proprioception
- (B) Fine touch, vibration, and two-point discrimination
- (C) Facial pain only
- (D) Emotional and visceral responses to pain
- (E) Precise location and intensity of the stimulus

5 A 70-year-old man on high-intensity atorvastatin after a large vessel stroke is treated for pneumonia with clarithromycin. He develops muscle pain and dark urine.

Which mechanism explains this?

- (A) Clarithromycin inhibits CYP3A4, raising atorvastatin levels
- (B) Atorvastatin inhibits clarithromycin metabolism
- (C) Clarithromycin directly damages skeletal muscle
- (D) Both drugs prolong the QT interval
- (E) Clarithromycin induces HMG-CoA reductase in skeletal muscle cells

6 A 27-year-old woman has had recurrent left-sided throbbing headaches since her teens, each lasting about a day, with nausea and a wish to lie in a dark, quiet room. Walking up stairs makes them worse. Examination between attacks is normal.

What is the most likely diagnosis?

- (A) Migraine without aura
- (B) Idiopathic intracranial hypertension
- (C) Cluster headache
- (D) Subarachnoid hemorrhage
- (E) Tension-type headache

7 A 64-year-old woman meets criteria for probable corticobasal syndrome, with asymmetric limb rigidity, dystonia, limb apraxia, and alien limb.

Which statement about her underlying pathology is correct?

- (A) DAT SPECT confirms CBD pathology and excludes PSP
- (B) It is always caused by α-synuclein in oligodendrocytes
- (C) PSP, Alzheimer disease, or other pathologies can cause it
- (D) It is always caused by CBD
- (E) MRI shows the hot cross bun sign

8 A 64-year-old woman with acute angle closure has an IOP of 62 mmHg. She has received acetazolamide, timolol, and apraclonidine.

When should pilocarpine be added?

(A) Once IOP is below 40 mmHg
(B) Only if IV mannitol fails
(C) Immediately, as the first drug
(D) Never, because it worsens angle closure
(E) Only after laser iridotomy

9 A 59-year-old man has a minor ischemic stroke with mild hand weakness. He was not given thrombolysis, and there is no cardioembolic source.

Which antiplatelet regimen is most appropriate?

(A) No antiplatelet until day 14
(B) Aspirin plus clopidogrel for 21 days, then one antiplatelet
(C) Clopidogrel plus warfarin for 3 months
(D) Aspirin alone indefinitely
(E) Aspirin plus apixaban for 21 days, then apixaban alone for life

10 In Parkinson disease, SNc dopamine neurons die.

What happens to GPi output?

(A) It is unchanged
(B) It becomes excitatory
(C) It increases, inhibiting the thalamus and causing bradykinesia
(D) It stops entirely, causing ballismus
(E) It decreases, releasing the VA and VL thalamus and causing chorea

11 A 35-year-old man who smokes wakes at 2 am most nights for 3 weeks with excruciating pain behind his right eye lasting 45 minutes. He paces the room, and his right eye waters and his right nostril runs.

What is the most likely diagnosis?

(A) Migraine with aura
(B) Tension-type headache
(C) Acute angle-closure glaucoma
(D) Cluster headache
(E) Trigeminal neuralgia

12 A 74-year-old man with DLB has troublesome visual hallucinations and worsening attention.

Which drug is most appropriate first?

(A) Rivastigmine
(B) Amitriptyline
(C) High-dose levodopa
(D) Haloperidol
(E) Risperidone

13 A neurophysiologist records inhibitory transmission onto motor neurons in the spinal cord.

Which transmitter is inhibitory mainly in the cord?

(A) Glutamate
(B) Dopamine
(C) Acetylcholine
(D) Norepinephrine
(E) Glycine

14 A patient with a large hematoma shows signs of impending herniation while awaiting surgery.

Which measure lowers ICP fastest as a bridge?

(A) Long-term hyperventilation for days
(B) Lumbar puncture
(C) High-dose oral corticosteroids
(D) Lying the patient flat
(E) Mannitol or hypertonic saline

15 A physiology student rides a lift that accelerates upward.

Which structure senses this vertical linear acceleration?

(A) Saccule
(B) Horizontal canal
(C) Utricle
(D) Posterior canal
(E) Cochlea

16 A 79-year-old man taking dabigatran for A-fib has a left lobar hemorrhage on CT. He took his last dose 3 hours ago.

Which reversal agent is most appropriate?

(A) Protamine
(B) Idarucizumab
(C) IV vitamin K
(D) Cryoprecipitate
(E) Andexanet alfa

---

17 A 50-year-old man has right facial weakness after temporal bone surgery, with a dry right eye but normal taste and hearing.

Which branch's fibers explain the dry eye?

(A) Greater petrosal nerve to the lacrimal gland
(B) Ophthalmic division of CN V
(C) Branches within the parotid gland
(D) Nerve to stapedius
(E) Chorda tympani to the submandibular and sublingual glands

---

18 In early Huntington disease, indirect-pathway striatal neurons die first.

What happens to GPi output and movement?

(A) GPi inhibition rises, causing bradykinesia
(B) GPi inhibition falls, causing chorea
(C) STN excitation rises, causing rigidity
(D) Thalamic output falls, causing akinesia
(E) Dopamine release falls, causing tremor

---

19 A 57-year-old man with a large ICH becomes drowsy, and his ICP is raised. Mannitol is given.

How does mannitol lower ICP?

(A) It blocks calcium entry into cerebral arteries
(B) It constricts cerebral arteries, lowering blood volume
(C) It lowers systemic BP to below 140
(D) It reduces CSF production by the choroid plexus
(E) It raises plasma osmolality, drawing water out of the brain

---

20 A 6-week-old infant has a white pupil in the right eye on a photograph. The red reflex is absent on that side.

Why is early treatment important?

(A) To prevent retinal detachment
(B) To prevent amblyopia
(C) Because the lens will otherwise swell and close the angle
(D) To prevent phacolytic glaucoma
(E) Because congenital cataracts become malignant

---

21 A 68-year-old man with hypertension and glaucoma has sudden painless blurring in his left eye. The fundus shows dilated, tortuous veins with hemorrhages in all four quadrants.

What is the most likely diagnosis?

(A) Retinal detachment
(B) Wet macular degeneration
(C) CRAO
(D) Anterior ischemic optic neuropathy
(E) CRVO

---

22 Touching the left cornea causes neither eye to blink, whereas touching the right cornea causes both eyes to blink.

Which nerve is damaged?

(A) Left CN III
(B) Right V1
(C) Right CN VII
(D) Left V1
(E) Left CN VII

---

23 A 60-year-old man with 12 years of Parkinson disease has disabling motor fluctuations and dyskinesia despite optimized levodopa, entacapone, and an agonist. His cognition is normal.

Which treatment is most appropriate?

(A) Adding haloperidol
(B) Tetrabenazine
(C) DBS
(D) Stopping levodopa
(E) Adding a second dopamine agonist

---

24 A 67-year-old man arrives 50 minutes after sudden right face and arm weakness and difficulty speaking. His BP is 176/98 mm Hg.

Which of the following should be done first?

(A) MRI with diffusion-weighted imaging
(B) Glucose and non-contrast CT head
(C) Aspirin 300 mg
(D) Lower his BP to below 140/90
(E) IV alteplase

25 A 30-year-old man has orofacial chorea that worsens when he eats, tongue biting, and raised CK. His HTT repeat length is normal, and his blood film shows spiky red cells. His parents are unaffected.

What is the most likely diagnosis?

(A) Hemiballismus from a stroke
(B) Juvenile Huntington disease
(C) Chorea-acanthocytosis
(D) Spinocerebellar ataxia
(E) Wilson disease

26 A 78-year-old woman with progressive memory loss has a headache and left homonymous hemianopia. CT shows a right occipital lobar hemorrhage. MRI with SWI shows many lobar microbleeds. She has no history of hypertension.

What is the most likely cause?

(A) Cerebral venous sinus thrombosis
(B) Arteriovenous malformation
(C) Hypertension
(D) Cerebral amyloid angiopathy
(E) Mycotic aneurysm

27 A 50-year-old woman has constant vertigo and vomiting for 1 day, with no other neurologic signs. The HINTS examination is reassuring.

What does this suggest?

(A) Peripheral vertigo, with MRI if still unclear
(B) Multiple sclerosis needing gadolinium MRI
(C) Vestibular migraine needing a triptan
(D) Posterior circulation stroke needing thrombolysis
(E) Cerebellar hemorrhage needing surgery

28 A 74-year-old woman has a new headache over her temples for 3 weeks, with pain in her jaw when chewing. Her ESR is 88 mm/h.

Which step is most appropriate?

(A) 100% oxygen during attacks
(B) Prompt high-dose glucocorticoids
(C) Non-contrast CT, then a lumbar puncture
(D) Stop regular analgesics
(E) Triptan at headache onset

29 A 49-year-old woman with aneurysmal SAH is started on oral nimodipine for 21 days.

What is the main purpose of nimodipine?

(A) Preventing rebleeding before repair
(B) Preventing seizures after the bleed
(C) Lowering ICP from hydrocephalus
(D) Preventing delayed ischemia from vasospasm
(E) Lowering BP until the aneurysm has been secured by coiling

30 A woman thought to have NMOSD has CSF oligoclonal bands, short peripheral cord lesions, and periventricular lesions at right angles to the ventricles.

Which diagnosis do these findings favor?

(A) MOGAD
(B) Multiple sclerosis
(C) Spinal cord infarction
(D) Area postrema syndrome
(E) AQP4-IgG-positive NMOSD

31 A 62-year-old smoker has proximal leg weakness, dry mouth, and reduced reflexes that briefly improve after repeated contraction. CXR shows a hilar mass.

What is the target of the autoantibody?

(A) Acetylcholinesterase
(B) Voltage-gated sodium channels
(C) Muscle-specific kinase
(D) Postsynaptic nicotinic acetylcholine receptors
(E) Presynaptic voltage-gated calcium channels

32 A 66-year-old vegan woman stomps when she walks and is worse in the dark. She stands steadily with her eyes open but sways and nearly falls when she closes them.

Which pathway is damaged?

- (A) Cerebellar vermis
- (B) Corticospinal tracts
- (C) Basal ganglia
- (D) Right cerebellar hemisphere
- (E) Dorsal columns

33 A 30-year-old woman with highly active relapsing MS is to start natalizumab.

Which test should be checked first?

- (A) Visual fields
- (B) JC virus antibody status
- (C) TPMT activity
- (D) Anti-IgA antibodies
- (E) HLA-B*1502 genotype

34 Non-contrast CT 3 hours after sudden aphasia and right hemiparesis is read as showing loss of the gray-white border over the left insula.

What does this finding represent?

- (A) A normal variant
- (B) An old lacunar infarct
- (C) Calcification of the basal ganglia
- (D) Subarachnoid hemorrhage
- (E) An early sign of ischemia

35 A 60-year-old hyperopic woman with a shallow anterior chamber takes an over-the-counter cold remedy containing pseudoephedrine. That night she develops a painful red eye with halos.

How did the remedy trigger her symptoms?

- (A) It dilated the pupil, letting the bunched iris narrow the angle
- (B) It constricted the pupil, causing pupillary block against the lens
- (C) It clogged the trabecular meshwork with protein
- (D) It swelled the lens through sorbitol
- (E) It decreased aqueous production

36 A 45-year-old man has bilateral pressing headaches on 18 days a month for a year. Neurologic examination is normal, and he uses ibuprofen often.

Which drug is best for prevention?

- (A) Prednisone
- (B) Verapamil
- (C) Lithium
- (D) Amitriptyline
- (E) Sumatriptan nasal spray

37 A 63-year-old man has mild memory problems, and the diagnosis is uncertain after clinical assessment and MRI.

Which CSF pattern supports Alzheimer disease?

- (A) Increased Aβ42 with decreased tau and phosphorylated tau
- (B) Oligoclonal bands with raised white cells
- (C) Decreased Aβ42 with increased tau and phosphorylated tau
- (D) Raised 14-3-3 protein only
- (E) Normal Aβ42 and normal tau

38 A 59-year-old man with diabetes has burning pain in both feet that disturbs his sleep. He also has depression.

Which drug treats both, by strengthening descending inhibition?

- (A) Paracetamol
- (B) Ketamine
- (C) Duloxetine
- (D) Ibuprofen
- (E) Morphine

39 A 41-year-old man woke 2 days ago with severe hearing loss in his left ear. Otoscopy is normal, and audiometry shows a 50 dB sensorineural loss across 4 consecutive frequencies.

Which is the most appropriate treatment?

- (A) Oral amoxicillin
- (B) Microsuction
- (C) A hearing aid fitted once his hearing stabilizes in 3 months
- (D) Oral prednisone 1 mg/kg for 7 days, then tapered
- (E) Observation for 3 months

40 A 31-year-old woman with relapsing-remitting MS has 5 days of new numbness and weakness of her left leg. Infection has been excluded.

Which is the most appropriate treatment?

(A) Plasma exchange first
(B) IVIG
(C) Start natalizumab today
(D) IV methylprednisolone for 3–7 days
(E) Baclofen

---

41 A 49-year-old woman had a ruptured aneurysm coiled 6 days ago. Today she becomes confused and develops new right arm weakness. CT head shows no new bleeding.

Which complication is most likely?

(A) Hemorrhagic transformation
(B) Central post-stroke pain
(C) Malignant MCA infarct from the initial bleed
(D) Rebleeding
(E) Vasospasm and delayed cerebral ischemia

---

42 A 61-year-old woman speaks fluently but cannot name common objects and asks "what is a fork?" when told to pick one up.

Atrophy of which region is most likely?

(A) Right anterior temporal lobe
(B) Left inferior frontal lobe and insula
(C) Left anterior temporal lobe
(D) Occipital cortex
(E) Bilateral hippocampi

---

43 A 27-year-old woman has episodes that begin with a rising feeling in her stomach and intense déjà vu, followed by staring, chewing movements, and no memory of the event. Afterward she is confused for 20 minutes.

Where do these seizures most likely start?

(A) Temporal lobe
(B) Frontal lobe motor cortex
(C) Occipital lobe
(D) Cerebellum
(E) Both hemispheres at once

---

44 On day 5 after aneurysmal SAH, a 56-year-old woman's sodium falls to 127 mmol/L.

Which is the most appropriate approach?

(A) Give a loop diuretic
(B) Strict fluid restriction to 800 mL a day until sodium recovers
(C) Treat the cause and avoid fluid restriction
(D) Stop nimodipine
(E) Hypotonic IV fluids

---

45 A 70-year-old man falls forward and hits his chin, extending his neck. He has weakness of both arms, worst in his hands, with only mild leg weakness, and loss of pain and temperature sensation over his shoulders.

Which cord syndrome is this?

(A) Complete cord transection
(B) Posterior cord syndrome
(C) Central cord syndrome
(D) Anterior cord syndrome
(E) Brown-Séquard syndrome

---

46 A 36-year-old woman thought to have MS worsened after starting interferon beta. Testing now shows AQP4-IgG, and her diagnosis is changed to NMOSD.

Which relapse-preventing drug is most appropriate?

(A) Glatiramer acetate
(B) Natalizumab
(C) Rituximab
(D) Dimethyl fumarate
(E) Fingolimod

---

47 A 59-year-old man uses a once-daily drop for open-angle glaucoma. Months later, his treated eye has a darker iris and longer eyelashes.

How does this drug lower IOP?

(A) It increases uveoscleral outflow
(B) It inhibits carbonic anhydrase
(C) It constricts the pupil
(D) It decreases aqueous production by the ciliary body
(E) It blocks ciliary β receptors

48 A 33-year-old woman with MS notices that her leg weakness and blurred vision worsen after a hot shower and settle within an hour.

What is this phenomenon?

- (A) Lhermitte sign
- (B) An acute relapse
- (C) Charcot triad
- (D) Secondary progression
- (E) Uhthoff phenomenon

---

49 A 70-year-old man has sudden left hemiplegia and neglect. His eyes are deviated to the right.

Where is the lesion?

- (A) Left hemisphere
- (B) Left pontine gaze center
- (C) Right medial longitudinal fasciculus
- (D) Right hemisphere
- (E) Dorsal midbrain

---

50 A 20-year-old man with an epidural hematoma on CT has a falling GCS, a dilated left pupil, and midline shift.

What is the most appropriate management?

- (A) Burr hole drainage next week
- (B) Observation with repeat CT in 24 hours
- (C) Oral corticosteroids
- (D) Lumbar puncture to lower pressure
- (E) Emergency craniotomy and evacuation

TEST A

Answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	11 A B C D E	21 A B C D E	31 A B C D E	41 A B C D E
2 A B C D E	12 A B C D E	22 A B C D E	32 A B C D E	42 A B C D E
3 A B C D E	13 A B C D E	23 A B C D E	33 A B C D E	43 A B C D E
4 A B C D E	14 A B C D E	24 A B C D E	34 A B C D E	44 A B C D E
5 A B C D E	15 A B C D E	25 A B C D E	35 A B C D E	45 A B C D E
6 A B C D E	16 A B C D E	26 A B C D E	36 A B C D E	46 A B C D E
7 A B C D E	17 A B C D E	27 A B C D E	37 A B C D E	47 A B C D E
8 A B C D E	18 A B C D E	28 A B C D E	38 A B C D E	48 A B C D E
9 A B C D E	19 A B C D E	29 A B C D E	39 A B C D E	49 A B C D E
10 A B C D E	20 A B C D E	30 A B C D E	40 A B C D E	50 A B C D E

Fill one bubble for each question. The answer key at the start of the Test A answers uses the same layout.

Test B

50 questions | 60 minutes | Start the timer

1 A 45-year-old woman with Huntington disease starts the first-line drug for chorea. Her chorea improves, but she becomes withdrawn and tearful.

How does this drug act?

- (A) It blocks VMAT2, depleting vesicular dopamine
- (B) It enhances GABA-A receptors
- (C) It inhibits dopamine synthesis
- (D) It blocks D2 receptors on striatal medium spiny neurons
- (E) It blocks NMDA receptors

2 A 70-year-old man with a COPD exacerbation is given high-flow oxygen, and his $PaCO_2$ rises.

Which explanation is most accepted?

- (A) Oxygen paralyzes the diaphragm
- (B) Ventilation-perfusion mismatch and the Haldane effect
- (C) Loss of his hypoxic drive makes him stop breathing for long periods
- (D) Oxygen blocks the central chemoreceptors on the ventral medulla
- (E) Oxygen stimulates the apneustic center

3 A 76-year-old man has a symptomatic chronic subdural hematoma with mild midline shift. His anticoagulant has been reversed.

Which procedure is usually used?

- (A) Craniectomy for a cerebellar bleed
- (B) Endovascular coiling
- (C) Burr hole drainage
- (D) Lumbar drain
- (E) Stereotactic radiosurgery

4 A student traces unconscious proprioception from leg muscle spindles to the cerebellum through the uncrossed dorsal tract.

Where does this tract arise, and which peduncle does it enter?

- (A) Red nucleus, inferior cerebellar peduncle
- (B) Spinal border cells, superior cerebellar peduncle
- (C) Dorsal horn, superior cerebellar peduncle
- (D) Clarke nucleus, inferior cerebellar peduncle
- (E) Nucleus gracilis, middle cerebellar peduncle

5 An 86-year-old woman has late Alzheimer disease with mutism, incontinence, and difficulty swallowing.

What is the most common cause of death in Alzheimer disease?

- (A) Myocardial infarction
- (B) Pneumonia
- (C) Status epilepticus
- (D) Hepatic failure
- (E) Lobar ICH

6 A 62-year-old East Asian woman has sudden severe pain in her right eye, halos around lights, headache, and vomiting after an evening in a dark cinema. The eye is red and firm, the cornea hazy, and the pupil fixed and mid-dilated. IOP is 58 mmHg.

After lowering the IOP medically, which treatment is definitive?

- (A) Topical latanoprost long term
- (B) Topical corticosteroids
- (C) Topical atropine
- (D) Laser trabeculoplasty
- (E) Laser peripheral iridotomy

7 Fetal MRI shows an enlarged posterior fossa, absence of the cerebellar vermis, and a cystic fourth ventricle, with dilated lateral ventricles.

Which malformation is this?

- (A) Holoprosencephaly
- (B) Myelomeningocele
- (C) Dandy-Walker malformation
- (D) Chiari 2 malformation with myelomeningocele
- (E) Chiari 1 malformation

8 A 50-year-old man with an expanding right temporal hematoma develops a dilated, unreactive right pupil and a right eye that rests down and out.

Which herniation and structure explain this?

- (A) Tonsillar herniation compressing the medulla
- (B) Central herniation stretching pontine arteries
- (C) Uncal herniation compressing CN III
- (D) Subfalcine herniation compressing the ACA
- (E) Upward herniation distorting the midbrain

9 A student notes that smell differs from the other senses in its route to the cortex.

What is distinctive about it?

- (A) It crosses entirely at the chiasm
- (B) Its receptor neurons never regenerate once they are lost
- (C) It is carried by the trigeminal nerve
- (D) It reaches cortex without a thalamic relay
- (E) It relays in the medial geniculate nucleus before the cortex

10 A neurologist notes that her patient with DLB draws a clock poorly and loses track of conversations, yet recalls recent events well.

Which feature of DLB explains the relatively spared memory?

- (A) Lewy bodies in the occipital cortex
- (B) Relative preservation of the medial temporal lobe
- (C) Lewy bodies in the dorsal motor nucleus of the vagus
- (D) Dopamine loss from the substantia nigra
- (E) Loss of the nucleus basalis of Meynert

11 A 34-year-old woman has a stroke the day after a long-haul flight, during which her calf became swollen. Workup finds no A-fib or carotid disease, and a bubble echocardiogram shows a patent foramen ovale.

Which mechanism best explains her stroke?

- (A) Global hypoperfusion causing a watershed infarct in the cortex
- (B) A venous thrombus crossing a right-to-left shunt
- (C) Lipohyalinosis of penetrating arteries
- (D) A left atrial thrombus from A-fib
- (E) Atherosclerotic thrombus in the ICA

12 A 52-year-old woman has a tremor of both hands when holding a cup or writing, which improves with a glass of wine. Her father had the same tremor. There is no bradykinesia.

Which drug is most appropriate?

- (A) Trihexyphenidyl
- (B) Propranolol
- (C) Haloperidol
- (D) Levodopa
- (E) Tetrabenazine

13 A 42-year-old man has restless, flowing, involuntary movements of his fingers, face, and trunk, with irritability and poor planning.

Loss of which neurons first explains his movements?

- (A) Indirect-pathway medium spiny neurons projecting to the GPe
- (B) Direct-pathway D1 medium spiny neurons projecting to the GPi
- (C) Dopamine neurons of the SNc
- (D) Cholinergic neurons of the nucleus basalis
- (E) Purkinje cells of the cerebellum

14 A 78-year-old woman admitted with a right MCA stroke is hungry and asks for breakfast. She has mild dysarthria.

What should happen before she eats or drinks?

- (A) Carotid duplex
- (B) Bubble echocardiogram
- (C) A swallow screen
- (D) EEG
- (E) Repeat CT head

15 A 24-year-old woman who sleeps in her contact lenses has 2 days of a painful red right eye, photophobia, and blurred vision. Slit lamp examination shows a white stromal infiltrate with an overlying epithelial defect.

Which is the most appropriate next step?

- (A) Topical corticosteroids
- (B) Corneal scraping, then urgent topical antibiotics
- (C) Oral acetazolamide
- (D) Artificial tears and cool compresses, with review in a week
- (E) Topical antihistamine drops

16 An 82-year-old man has a supratentorial hemorrhage estimated at 40 mL on CT, with intraventricular extension. His GCS is 9.

What is his ICH score?

- (A) 2
- (B) 5
- (C) 4
- (D) 6
- (E) 3

17 A 76-year-old man struggles to follow conversation in restaurants and mishears words like "fit" and "sit." Both eardrums are normal.

Which audiogram pattern is most likely?

- (A) A notch at 4 kHz with recovery at 8 kHz
- (B) An air-bone gap in both ears
- (C) Bilateral, high-frequency, downsloping SNHL
- (D) Low-frequency SNHL in one ear that fluctuates
- (E) A flat loss in one ear with an air-bone gap in the other

18 A newborn of hearing parents fails the hearing screen and has severe bilateral SNHL with no other abnormalities. Congenital CMV testing is negative.

Mutations in which gene are most likely?

- (A) EDNRB
- (B) PAX3
- (C) MITF
- (D) GJB2
- (E) The merlin gene

19 Four months after a stroke, a 63-year-old man has burning pain over his whole right side, worsened by light touch from his clothes. Examination shows reduced sensation on the right but no new deficit.

Which earlier infarct site is most likely?

- (A) Left thalamus
- (B) Right cerebellar hemisphere
- (C) Left frontal eye field
- (D) Left occipital cortex
- (E) Left caudate

20 A 45-year-old man has vestibular neuritis with severe vertigo and nausea. He is given meclizine and asks to keep taking it for a month.

Why is it limited to about 3 days?

- (A) It damages vestibular hair cells
- (B) It causes tolerance within 3 days
- (C) It raises the risk of BPPV
- (D) It delays central compensation
- (E) It causes permanent hearing loss

21 After a stab wound to the right side of the back at T8, a man has weakness and loss of vibration and position sense in the right leg.

Where else is sensation lost?

- (A) All sensation in both legs
- (B) Pain and temperature in both arms
- (C) Vibration in the left leg
- (D) Pain and temperature in the right leg
- (E) Pain and temperature in the left leg

22 A 45-year-old woman had relapsing-remitting MS from age 30. Over the last 3 years her walking has gradually worsened, without any relapses.

Which course does this describe?

- (A) Clinically isolated syndrome
- (B) Radiologically isolated syndrome
- (C) Relapsing-remitting
- (D) Primary progressive
- (E) Secondary progressive

23 Two days after SAH, a 60-year-old man becomes increasingly drowsy. CT shows enlarged lateral ventricles.

What is the mechanism of this complication?

- (A) The aneurysm compresses the cerebral aqueduct from the start
- (B) Hyponatremia causes cerebral edema
- (C) Vasospasm infarcts the periventricular white matter
- (D) Blood stimulates the choroid plexus to make more CSF
- (E) Blood blocks CSF absorption at the arachnoid granulations

24 A 38-year-old man with flank pain and hematuria has enlarged kidneys full of cysts on ultrasound. His mother died suddenly of a brain hemorrhage at 45.

Which type of brain hemorrhage is he at increased risk of?

- (A) Subarachnoid hemorrhage from a saccular aneurysm
- (B) Venous infarction from sinus thrombosis
- (C) Subdural hemorrhage from bridging veins
- (D) Deep hemorrhage from Charcot-Bouchard aneurysms in the pons
- (E) Lobar hemorrhage from amyloid angiopathy

25 Massage over the carotid sinus slows a patient's heart rate.

Which nerve carries the afferent signal?

- (A) CN IX
- (B) CN VII
- (C) CN X
- (D) CN XII
- (E) CN XI

26 A 45-year-old woman with NMOSD is starting RTX every 6 months.

Which monitoring is needed?

- (A) Visual evoked potentials yearly
- (B) Serum AQP4-IgG titer before each dose to set the dose
- (C) JC virus status every 6 months
- (D) Lymphocyte count to keep it above 0.2 × 10^9/L
- (E) Immunoglobulin levels before treatment and yearly

27 A 76-year-old man with hypertension, diabetes, and two TIAs has had a stepwise decline in thinking over 2 years, with sudden drops followed by plateaus.

Which diagnosis is most likely?

- (A) Alzheimer disease
- (B) NPH
- (C) Prion disease
- (D) Vascular dementia
- (E) Frontotemporal dementia

28 A neurologist proposes testing MOG-IgG in every patient with possible MS.

Why is this discouraged?

- (A) The assay requires a brain biopsy
- (B) MOGAD is treated the same as MS, so the result changes nothing
- (C) MOG-IgG is positive in most MS
- (D) MOG-IgG predicts disability so strongly that it would alarm patients
- (E) MOG-IgG is found in only 0.4% of MS, so false positives result

29 Epidural hematomas are much more common in adolescents and young adults than in older adults.

What best explains this?

- (A) Bridging veins are more stretched
- (B) Arteries are more atherosclerotic
- (C) The dura is less firmly attached to the skull
- (D) Sutures are already fused
- (E) The brain is more atrophic

30 A 45-year-old woman with lifelong migraine has the worst headache of her life, reaching full intensity within a minute while lifting a box.

What is the most appropriate next step?

- (A) Non-contrast CT of the head
- (B) Reassure that this is her usual migraine
- (C) ESR and temporal artery biopsy
- (D) Start propranolol
- (E) Sumatriptan and review in a week

31 A 60-year-old woman has lost the left half of the visual field in both eyes, without macular sparing.

Which lesion fits?

- (A) Right optic nerve
- (B) Left Meyer loop
- (C) Right optic tract
- (D) Left optic tract
- (E) Optic chiasm

32 An 84-year-old woman became confused over 2 days. Her attention wanders, she is worse at night, and she has a urinary tract infection.

What is the most likely cause of her confusion?

- (A) Dementia with Lewy bodies
- (B) Delirium
- (C) Depression
- (D) Alzheimer disease
- (E) Frontotemporal dementia

33 A 24-year-old woman with poorly controlled type 1 diabetes has blurred vision. Slit lamp examination shows white, flake-like cortical opacities in both lenses.

Which mechanism explains these opacities?

- (A) Urochrome pigment collects in the nucleus
- (B) VEGF drives new vessels from the iris into the lens
- (C) Macrophages clog the trabecular meshwork
- (D) Sorbitol accumulates and draws water into the lens
- (E) Lens epithelium undergoes fibrous metaplasia at the back of the lens

34 A 70-year-old woman sees two overlapping images when she looks at the moon with her right eye alone. Covering the right eye stops it.

What is the most likely cause?

- (A) Right optic neuritis
- (B) Right CN VI palsy from microvascular ischemia
- (C) Cataract in the right eye
- (D) Left INO
- (E) Ocular myasthenia gravis

35 A 70-year-old woman with late-onset Alzheimer disease carries two copies of the APOE ε4 allele.

How does APOE ε4 raise her risk?

- (A) It forms part of γ-secretase
- (B) It increases APP gene dosage
- (C) It blocks α-secretase
- (D) It hyperphosphorylates tau directly
- (E) It slows Aβ clearance

36 A 20-month-old boy has a 2-minute generalized seizure during a fever of 39.2 °C from an ear infection. He recovers fully within 30 minutes and has no meningeal signs.

How is this seizure best classified?

- (A) Absence seizure
- (B) Status epilepticus
- (C) Epilepsy needing long-term medication
- (D) Simple febrile seizure
- (E) Complex febrile seizure

37 A 48-year-old man has hoarseness, difficulty swallowing, and weakness turning his head to the left and shrugging his right shoulder.

Where is the lesion?

- (A) Left medulla
- (B) Right jugular foramen
- (C) Right cerebellopontine angle
- (D) Right C5 root
- (E) Right cavernous sinus

38 A 66-year-old man with MSA faints on standing. He starts a short-acting α1 agonist prodrug, taken during the day only.

Why is it not taken at night?

- (A) It is inactivated while lying down
- (B) It causes nocturnal bradycardia
- (C) It worsens REM sleep behavior disorder
- (D) It causes nighttime hypoglycemia
- (E) It causes supine hypertension

39 A 40-year-old man presents during a typical attack of cluster headache.

What is the most appropriate acute treatment?

- (A) Oral sumatriptan
- (B) Oral verapamil
- (C) Oral prednisone for 5 days
- (D) 100% oxygen by face mask
- (E) Oral ibuprofen

40 A 67-year-old man with Parkinson disease starts levodopa combined with carbidopa.

What is the role of carbidopa?

- (A) It stimulates striatal dopamine receptors directly
- (B) It blocks dopamine breakdown by MAO-B in the brain
- (C) It helps levodopa cross the BBB on the amino acid transporter
- (D) It blocks COMT in the gut and liver
- (E) It blocks peripheral DDC, so more levodopa reaches the brain

41 A student tests the gag reflex.

Which nerves form its afferent and efferent limbs?

- (A) CN X afferent, CN IX efferent
- (B) CN IX afferent, CN XII efferent
- (C) CN X afferent, CN XI efferent
- (D) CN V afferent, CN VII efferent
- (E) CN IX afferent, CN X efferent

42 A 41-year-old woman with MS and severe spasticity runs out of her antispasticity drug. Two days later, she is confused, febrile, and has a seizure.

Which drug was most likely stopped?

- (A) Baclofen
- (B) Fingolimod
- (C) Oxybutynin
- (D) Interferon beta
- (E) Gabapentin

43 During a deep breath, slowly adapting stretch receptors in the airways fire increasingly.

What reflex do they mediate?

- (A) Reflex bradycardia through the carotid sinus nerve
- (B) Inhibition of inspiration to prevent overinflation
- (C) Bronchoconstriction
- (D) Cough
- (E) Faster, shallower breathing in response to interstitial edema

44 Two patients have recurrent headaches. One has bilateral pressing pain that walking does not worsen; the other has unilateral throbbing pain worsened by activity, with nausea.

Which feature best separates the first patient's headache from migraine?

- (A) Pain is not worsened by routine activity
- (B) Pain responds to NSAIDs within an hour or two
- (C) Pain is moderate in intensity
- (D) Pain recurs over years
- (E) Pain lasts several hours

45 A 58-year-old man with bvFTD is started on donepezil by a doctor who assumes it works as in Alzheimer disease.

What is the expected effect?

- (A) Prevention of ALS in C9orf72 carriers
- (B) Slowed progression of frontal and temporal atrophy
- (C) No proven benefit, and behavior may worsen
- (D) Improved behavior and cognition within weeks
- (E) Reduced hyperorality

46 A 25-year-old man with drug-resistant focal seizures is being assessed for surgery. MRI is normal.

Which finding on ictal SPECT, injected within 90 seconds of onset, marks the seizure focus?

- (A) Normal symmetric perfusion
- (B) Increased perfusion at the focus
- (C) Reduced striatal uptake
- (D) Amyloid deposition at the focus
- (E) Decreased perfusion at the focus

47 A 28-year-old with epilepsy controlled on valproate plans a pregnancy next year.

Which advice best reduces the risk of a neural tube defect?

- (A) Start folic acid once pregnancy is confirmed at 8 weeks
- (B) Stop all antiseizure drugs once pregnant
- (C) Add vitamin B12 in the first trimester
- (D) Review the antiseizure drug and start folic acid before conception
- (E) Continue valproate unchanged and screen maternal serum AFP at 16 weeks

48 A 71-year-old woman with Parkinson disease has a drug added that prolongs the effect of each levodopa dose.

Which drug and mechanism fit?

- (A) Rasagiline, a COMT inhibitor
- (B) Trihexyphenidyl, a dopamine precursor
- (C) Amantadine, a D3 receptor agonist
- (D) Entacapone, a COMT inhibitor
- (E) Pramipexole, an MAO-B inhibitor

49 A 70-year-old man has parkinsonism that responded moderately to levodopa for several years, with later vertical gaze slowing. His neurologist expects a longer course than classic PSP.

Which PSP subtype fits?

- (A) PSP-P
- (B) PSP-CBS
- (C) PSP-PNFA
- (D) PSP-bvFTD
- (E) PSP-RS

50 A 50-year-old woman with vestibular neuritis has used meclizine and diazepam daily for 6 weeks and still feels dizzy.

Which is the most appropriate change?

- (A) Stop the suppressants and start vestibular rehabilitation
- (B) Perform the Epley maneuver weekly
- (C) Increase the diazepam dose
- (D) Add oral prednisone
- (E) Refer for intratympanic gentamicin to ablate the labyrinth

TEST B

Answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	11 A B C D E	21 A B C D E	31 A B C D E	41 A B C D E
2 A B C D E	12 A B C D E	22 A B C D E	32 A B C D E	42 A B C D E
3 A B C D E	13 A B C D E	23 A B C D E	33 A B C D E	43 A B C D E
4 A B C D E	14 A B C D E	24 A B C D E	34 A B C D E	44 A B C D E
5 A B C D E	15 A B C D E	25 A B C D E	35 A B C D E	45 A B C D E
6 A B C D E	16 A B C D E	26 A B C D E	36 A B C D E	46 A B C D E
7 A B C D E	17 A B C D E	27 A B C D E	37 A B C D E	47 A B C D E
8 A B C D E	18 A B C D E	28 A B C D E	38 A B C D E	48 A B C D E
9 A B C D E	19 A B C D E	29 A B C D E	39 A B C D E	49 A B C D E
10 A B C D E	20 A B C D E	30 A B C D E	40 A B C D E	50 A B C D E

Fill one bubble for each question. The answer key at the start of the Test B answers uses the same layout.

Test C

50 questions | 60 minutes | Start the timer

1 At autopsy, a patient with parkinsonism, early autonomic failure, and ataxia has α-synuclein inclusions in oligodendrocytes.

Which disease is this?

(A) Multiple system atrophy
(B) Typical Parkinson disease
(C) Progressive supranuclear palsy
(D) Corticobasal degeneration
(E) Frontotemporal lobar degeneration

2 An 84-year-old woman with brain atrophy develops a subdural hematoma after a minor bump to the head.

Which structures most likely bled?

(A) Bridging veins
(B) The vertebral artery
(C) The middle meningeal artery
(D) Charcot-Bouchard microaneurysms
(E) Berry aneurysm

3 A 60-year-old man has a small pontine stroke. Neither eye can look to the left, but convergence is preserved.

Which structure is damaged?

(A) Right MLF
(B) Left CN VI nucleus
(C) Left CN VI nerve
(D) Right frontal eye field
(E) Left CN III nucleus

4 A 60-year-old man no longer recognizes the faces of his friends and gets lost in his own neighborhood. MRI shows asymmetric atrophy.

Which region is most affected?

(A) Right anterior temporal lobe
(B) Left inferior frontal gyrus
(C) Cerebellar vermis
(D) Left anterior temporal lobe
(E) Dorsolateral prefrontal cortex

5 A student examines a sensory neuron whose cell body lies in a dorsal root ganglion, with one process that splits into a peripheral and a central branch.

Which type of neuron is this?

(A) Multipolar, like a motor neuron
(B) Pseudounipolar
(C) Interneuron
(D) Bipolar, with two separate processes
(E) Bipolar

6 A 39-year-old woman with AQP4-IgG-positive NMOSD is to start eculizumab.

Which step is required before treatment?

(A) HLA-B*1502 testing
(B) Checking TPMT activity alone
(C) Echocardiogram
(D) Meningococcal vaccination
(E) JC virus antibody testing

7 A 70-year-old man with early Alzheimer disease and confirmed amyloid is offered lecanemab. He carries two APOE4 alleles.

Which adverse effect is most increased by his genotype?

(A) Hyponatremia
(B) ARIA
(C) Bradycardia
(D) Hepatotoxicity
(E) Agranulocytosis

8 A 79-year-old woman with moderate to severe Alzheimer disease has memantine added to donepezil.

How does memantine act?

(A) It blocks dopamine reuptake
(B) It clears amyloid plaques
(C) It slows acetylcholine breakdown in the synapse
(D) It activates GABA-A receptors
(E) It blocks NMDA channels noncompetitively

9 A 30-year-old woman has a first attack of myelitis. CSF shows 85 white cells/μL with neutrophils and eosinophils and no oligoclonal bands.

Which diagnosis do these CSF findings favor?

- (A) Small vessel disease
- (B) Functional neurologic disorder
- (C) Multiple sclerosis
- (D) B12 deficiency
- (E) NMOSD

10 A 45-year-old man taking a long-standing antiseizure drug has swollen, overgrown gums. His dose was recently increased, and he now has nystagmus and an unsteady gait, with a level far higher than the dose increase predicted.

Which drug is he most likely taking?

- (A) Levetiracetam
- (B) Lamotrigine
- (C) Valproate
- (D) Phenytoin
- (E) Carbamazepine

11 A 7-month-old boy with infantile spasms and hypsarrhythmia is started on vigabatrin.

Which adverse effect requires monitoring?

- (A) Concentric visual field loss
- (B) Hepatotoxicity
- (C) Agranulocytosis
- (D) Gingival hyperplasia
- (E) Kidney stones

12 A tract from the medulla inhibits extensors, facilitates flexors, and contributes to breathing control.

Which tract is this?

- (A) Anterior corticospinal tract
- (B) Pontine reticulospinal tract, uncrossed
- (C) Medullary reticulospinal tract
- (D) Rubrospinal tract
- (E) Lateral vestibulospinal tract from the lateral nucleus

13 A 35-year-old man has 2 days of continuous vertigo after a cold, with hearing loss and ringing in his right ear. Nystagmus is unidirectional and suppressed by fixation.

What is the most likely diagnosis?

- (A) Labyrinthitis
- (B) BPPV
- (C) Vestibular neuritis
- (D) Ménière disease
- (E) Vestibular migraine

14 A neurosurgeon traces the lateral corticospinal tract from M1 toward the brainstem.

Through which part of the internal capsule does it pass?

- (A) Retrolenticular part
- (B) External capsule
- (C) Genu
- (D) Anterior limb
- (E) Posterior limb

15 A 61-year-old woman develops symmetric bradykinesia and rigidity weeks after starting a drug for chronic nausea. A DAT SPECT is normal.

Which drug is most likely responsible?

- (A) Metoclopramide
- (B) Scopolamine
- (C) Levodopa
- (D) Pramipexole
- (E) Ondansetron

16 A 50-year-old woman with Huntington disease has chorea that makes her drop things and fall. She has a history of depression.

If tetrabenazine is started, what must be monitored?

- (A) Serum lithium level
- (B) Liver enzymes weekly
- (C) Serum potassium
- (D) White cell count for agranulocytosis
- (E) Mood and suicidal thoughts

17 A 58-year-old man with 20 years of poorly controlled hypertension has a pontine hemorrhage.

Which vascular change most likely underlies the bleed?

- (A) Saccular aneurysm at a bifurcation of the circle of Willis
- (B) Charcot-Bouchard aneurysms in penetrating arterioles
- (C) A tangle of arteries draining directly into veins
- (D) Septic emboli lodging in distal arteries
- (E) Amyloid deposits in cortical vessel walls

18 A 40-year-old woman has twitching that starts in her left thumb, spreads over a minute to her hand and arm, then to the left side of her face.

What does this spread indicate?

- (A) A migraine aura in the occipital cortex
- (B) A seizure spreading along the motor cortex
- (C) A spinal cord lesion
- (D) Transient ischemia of the internal capsule
- (E) Peripheral nerve hyperexcitability

19 A 45-year-old man with epilepsy is found after a witnessed tonic-clonic seizure with a bitten tongue. An hour later he has left arm weakness that is improving. CT head is normal and glucose is normal.

Which is the most appropriate management of his weakness?

- (A) IV methylprednisolone
- (B) Start aspirin and clopidogrel
- (C) CT angiogram and mechanical thrombectomy
- (D) Observation, expecting it to resolve
- (E) IV alteplase

20 A 22-year-old woman has a head CT after a fall from a horse.

Which CT appearance indicates an epidural hematoma?

- (A) Round collection deep in the basal ganglia
- (B) Diffuse loss of gray-white matter differentiation
- (C) Biconvex collection that stops at the sutures
- (D) Blood in the basal cisterns and sulci
- (E) Crescent that crosses the sutures

21 A 41-year-old man has myelitis with a conus lesion. Serum MOG-IgG on a live cell-based assay is positive at 1:40, and AQP4-IgG is negative.

Under the 2023 criteria, can MOGAD be diagnosed?

- (A) No, myelitis is not a MOGAD syndrome
- (B) No, a titer of 1:100 or more is always required
- (C) Yes, without needing to exclude MS
- (D) No, MOG-IgG must also be found in CSF
- (E) Yes, the conus lesion is a supporting feature

22 A volume load stretches the atria of a healthy volunteer.

What response follows?

- (A) Increased sympathetic tone to the kidney
- (B) Reduced vagal tone to the heart
- (C) Lower renal sympathetic tone and ANP release, causing natriuresis
- (D) Vasopressin release driven by the area postrema
- (E) Higher RAAS activity and vasopressin release from the pituitary

23 During an attack of cluster headache, a man's right eyelid droops and his right pupil becomes small.

Which pathway explains these signs?

- (A) Parasympathetic overactivity at the iris
- (B) Reduced sympathetic supply to the eye
- (C) Compression of CN III
- (D) Optic nerve ischemia
- (E) Damage to CN VI

24 A 21-year-old woman collapsed after standing for an hour at a hot concert. She felt lightheaded and her vision greyed first, had a few brief jerks, and was fully alert within a minute.

Which tests are most appropriate?

- (A) Video EEG
- (B) MRI brain with gadolinium
- (C) ECG and orthostatic BP
- (D) DAT SPECT
- (E) Lumbar puncture

25 An 84-year-old woman in a nursing home became confused over 2 days. Her attention wanders, and she is drowsy in the morning but agitated at night. She has a fever and cloudy urine.

Which is the most likely cause of her confusion?

(A) Delirium
(B) Alzheimer disease
(C) Frontotemporal dementia
(D) Vascular dementia
(E) Normal pressure hydrocephalus

26 A student notes that CN IV is unusual among the cranial nerves.

Which feature makes it unusual?

(A) It runs free within the cavernous sinus rather than in its wall
(B) Every fiber crosses before leaving the back of the midbrain
(C) It carries parasympathetic fibers
(D) It leaves through the interpeduncular fossa beside the PComm
(E) Its nucleus is the horizontal gaze center

27 A 38-year-old office worker has headaches most afternoons, described as a tight band around her head. They are mild to moderate, do not stop her working, and come without nausea. Her neck and scalp muscles are tender.

What is the most likely diagnosis?

(A) Tension-type headache
(B) Trigeminal neuralgia
(C) Cluster headache
(D) Migraine without aura
(E) Chronic subdural hematoma

28 A 45-year-old man has had 3 attacks of vertigo lasting 2–4 hours, each with roaring tinnitus and fullness in his left ear. Audiometry shows left low-frequency sensorineural loss.

Which is the most appropriate first treatment?

(A) Surgical section of the vestibular nerve as first-line treatment
(B) Intratympanic gentamicin
(C) The Epley maneuver
(D) Less salt, caffeine, and alcohol, with a thiazide or betahistine
(E) Long-term daily meclizine

29 A 26-year-old woman with highly active relapsing MS is to start natalizumab.

Which test should be checked first?

(A) JC virus status
(B) AQP4 antibody
(C) Thiopurine methyltransferase
(D) HLA-B*1502
(E) First-dose ECG monitoring

30 Two patients after surgery have depressed breathing: one was given morphine, and the other propofol.

How do their CO2 response curves differ?

(A) Both shift it left
(B) Morphine flattens it; propofol shifts it right
(C) Morphine shifts it right; propofol flattens it
(D) Both leave it unchanged
(E) Morphine shifts it left; propofol shifts it right

31 A cardiac glycoside partially blocks the pump that maintains neuronal ion gradients.

What does this pump move with each cycle?

(A) 2 sodium out and 3 potassium in
(B) 1 calcium out and 3 sodium in
(C) 1 sodium in and 1 chloride in
(D) 3 sodium out and 2 potassium in
(E) 3 potassium out and 2 sodium in

32 A 62-year-old woman has horizontal double vision from a right abduction deficit, weakness of the whole right side of her face including the forehead, and weakness of her left arm and leg.

Which syndrome is this?

(A) Lateral pontine (AICA) syndrome
(B) Inferior medial pontine (Foville) syndrome
(C) Lateral medullary syndrome
(D) Ventral pontine (Millard-Gubler) syndrome
(E) Weber syndrome

33 A 70-year-old man notices his eyelashes have grown longer and his light brown iris has darkened since starting a glaucoma drop.

How does this drop lower IOP?

(A) Constricts ciliary body vessels through α1 receptors
(B) Reduces bicarbonate in the ciliary body
(C) Increases uveoscleral outflow
(D) Contracts the ciliary muscle to open the trabecular meshwork
(E) Decreases aqueous production by blocking ciliary β receptors

34 During a large regional block with lidocaine, a 39-year-old woman reports numbness around her mouth and then has a generalized seizure.

What is the most likely cause, and what follows if it progresses?

(A) Malignant hyperthermia, then rhabdomyolysis
(B) Anaphylaxis to an amide, then bronchospasm
(C) Vasovagal syncope, then bradycardia alone
(D) Opioid toxicity, then respiratory depression
(E) Systemic lidocaine toxicity, then arrhythmia and cardiac arrest

35 MRI of a woman with old MS plaques shows a lesion bright on DWI. The same lesion is bright on ADC.

What does this combination indicate?

(A) Acute infarct under 4.5 hours old
(B) An abscess
(C) Acute hemorrhage
(D) T2 shine-through, not acute infarct
(E) An active plaque that would enhance with gadolinium

36 A 4-year-old boy wakes with both eyelids stuck together by yellow discharge. His eyes are red, but his vision is normal.

Which treatment shortens this illness?

(A) Oral aciclovir
(B) Topical antihistamines and mast cell stabilizers
(C) Topical antibiotic drops such as chloramphenicol
(D) Topical corticosteroids to reduce the inflammation
(E) Intravitreal anti-VEGF

37 A 3-year-old boy has speech delay. Otoscopy shows dull, retracted eardrums with fluid behind them in both ears, persisting for 4 months. Audiometry shows conductive loss.

Which treatment is most appropriate?

(A) GJB2 testing alone
(B) Stapes surgery
(C) Oral prednisone
(D) Cochlear implants
(E) Ventilation tubes

38 A 40-year-old woman with migraine now has headaches on 20 days a month. She takes paracetamol or a triptan almost every day.

What is the most likely cause of the increase?

(A) Brain tumor
(B) Cluster headache
(C) Idiopathic intracranial hypertension
(D) Temporal arteritis
(E) Medication overuse headache

39 A 30-year-old man with a complete C6 spinal cord injury has a sudden pounding headache, flushing above the injury, and a BP of 210/120 mm Hg with HR 48/min.

Which is the most appropriate first step?

(A) Perform carotid sinus massage to slow the heart
(B) Give IV furosemide to lower his blood volume
(C) Check and relieve bladder distension
(D) Give atropine
(E) Lay him flat

40 An 80-year-old man with DLB has distressing psychosis despite rivastigmine, and his team decides an antipsychotic is essential.

Which choice is most appropriate?

- (A) Haloperidol at a low dose
- (B) Aripiprazole
- (C) Olanzapine
- (D) Low-dose quetiapine
- (E) Risperidone at night

41 A 74-year-old man suddenly develops violent, flinging movements of his left arm and leg.

Where is the lesion most likely?

- (A) Right cerebellar hemisphere
- (B) Left caudate
- (C) Right subthalamic nucleus
- (D) Right substantia nigra
- (E) Left subthalamic nucleus

42 A man with a large left subdural hematoma has a dilated left pupil and weakness of the left arm and leg.

Why is the weakness on the same side as the lesion?

- (A) The hematoma has crossed the midline
- (B) The opposite cerebral peduncle is pressed against the tentorium
- (C) The corticospinal tract does not decussate
- (D) CN III supplies the limbs
- (E) The medulla is compressed

43 An 83-year-old man with a hypermature cataract has a painful red eye with IOP of 44 mmHg. Gonioscopy shows an open angle, and white cells float in the anterior chamber.

What is the mechanism of his raised IOP?

- (A) A swollen, intumescent lens pushes the iris forward and closes the angle
- (B) Pupillary block from a dilated pupil
- (C) Neovascularization of the angle
- (D) Steroid-induced reduction in outflow through the meshwork
- (E) Lens proteins leak and macrophages clog the trabecular meshwork

44 A 60-year-old smoker has wasting and clawing of his left hand, numbness of his medial forearm, and a small left pupil with ptosis.

Which lesion best explains these findings?

- (A) Median nerve in the carpal tunnel
- (B) C8–T1 injury from an apical lung tumor
- (C) Ulnar nerve at the cubital tunnel
- (D) Left lateral medullary stroke from PICA occlusion
- (E) C5–C6 traction injury of the upper trunk

45 An 8-year-old girl has dozens of brief staring spells a day, with 3 Hz spike-and-wave discharges on EEG. She is started on the first-line drug.

Which target does it block?

- (A) GABA transaminase
- (B) T-type calcium channels
- (C) NMDA receptors
- (D) SV2A on vesicles
- (E) Voltage-gated sodium channels

46 A 70-year-old woman presents 7 hours after onset of left hemiparesis. CT shows no hemorrhage and CT angiogram shows no large vessel occlusion. She will not receive reperfusion therapy. Her BP is 196/104 mm Hg.

Which is the most appropriate approach to her BP?

- (A) Allow it up to 220/120
- (B) Lower it immediately to below 140/90
- (C) Give IV nicardipine to a systolic target of 140
- (D) Lower it to below 185/110 for thrombolysis
- (E) Give nimodipine

47 A 46-year-old man with an unsecured ruptured AComm aneurysm has BP 182/100 mm Hg while awaiting coiling.

Which BP target is most appropriate until the aneurysm is secured?

- (A) No treatment unless BP rises above 220/120 mm Hg
- (B) Systolic below 185 for thrombolysis
- (C) Systolic below 110 with nitroprusside
- (D) Below 130/80 mm Hg with oral agents started at once
- (E) Systolic below 160 with IV nicardipine or labetalol

48 A radiologist wants a sequence that makes lesions bright while keeping CSF dark, to show plaques next to the ventricles.

Which sequence is this?

- (A) T1 without contrast
- (B) DWI
- (C) MRA
- (D) FLAIR
- (E) SWI

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49 A 69-year-old woman who has taken levodopa-carbidopa for 6 years notices her tremor and slowness return about 2 hours before each dose is due.

Which is the most appropriate change?

- (A) Stop levodopa
- (B) Add trihexyphenidyl only
- (C) Add entacapone
- (D) Add haloperidol
- (E) Switch to amantadine alone

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50 A 70-year-old man with Parkinson disease starts levodopa combined with carbidopa.

What is the role of carbidopa?

- (A) It stimulates D3 receptors directly
- (B) It blocks NMDA receptors
- (C) It inhibits MAO-B in the brain
- (D) It blocks peripheral conversion of levodopa to dopamine
- (E) It inhibits COMT in the brain to prolong each dose

TEST C

Answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	11 A B C D E	21 A B C D E	31 A B C D E	41 A B C D E
2 A B C D E	12 A B C D E	22 A B C D E	32 A B C D E	42 A B C D E
3 A B C D E	13 A B C D E	23 A B C D E	33 A B C D E	43 A B C D E
4 A B C D E	14 A B C D E	24 A B C D E	34 A B C D E	44 A B C D E
5 A B C D E	15 A B C D E	25 A B C D E	35 A B C D E	45 A B C D E
6 A B C D E	16 A B C D E	26 A B C D E	36 A B C D E	46 A B C D E
7 A B C D E	17 A B C D E	27 A B C D E	37 A B C D E	47 A B C D E
8 A B C D E	18 A B C D E	28 A B C D E	38 A B C D E	48 A B C D E
9 A B C D E	19 A B C D E	29 A B C D E	39 A B C D E	49 A B C D E
10 A B C D E	20 A B C D E	30 A B C D E	40 A B C D E	50 A B C D E

Fill one bubble for each question. The answer key at the start of the Test C answers uses the same layout.

Test A answers

Answer key | Explanations

1 D	11 D	21 E	31 E	41 E
2 B	12 A	22 D	32 E	42 C
3 B	13 E	23 C	33 B	43 A
4 D	14 E	24 B	34 E	44 C
5 A	15 A	25 C	35 A	45 C
6 A	16 B	26 D	36 D	46 C
7 C	17 A	27 A	37 C	47 A
8 A	18 B	28 B	38 C	48 E
9 B	19 E	29 D	39 D	49 D
10 C	20 B	30 B	40 D	50 E

1 D MRI of the brain and orbits with gadolinium

Vision loss and red eye | Step 2

Optic neuritis causes painful monocular vision loss with an RAPD and reduced color vision, and is often the first sign of MS, so MRI of the brain and orbits is obtained. IV methylprednisolone speeds recovery without changing the outcome.

WHY NOT THE OTHERS

(E) Temporal artery biopsy is for GCA, after age 50.

2 B Subdural hematoma

Subdural hematoma | Step 1

A subdural hematoma is a crescent that crosses sutures but not the midline, because the falx stops it; it is bright when acute, isodense at 1–3 weeks, and dark when chronic.

3 B MOG-IgG

MOGAD | Step 2

Bilateral optic neuritis with disc edema, an optic nerve lesion over half its length, and perineural optic sheath enhancement are supporting features of MOGAD, which affects men and women equally.

WHY NOT THE OTHERS

(D) NMOSD favors posterior and chiasmal lesions, mostly in women.

4 D Emotional and visceral responses to pain

Physiology | Pain pathways and descending modulation | Step 1

The medial paleospinothalamic and archispinothalamic pathways project bilaterally to the reticular formation, PAG, and intralaminar thalamus, then to the cingulate, insula, and hypothalamus, carrying emotional and visceral responses.

WHY NOT THE OTHERS

(B) Fine touch travels in the dorsal columns.

(E) Location and intensity travel in the lateral tract.

5 A Clarithromycin inhibits CYP3A4, raising atorvastatin levels

Physiology | Clots and bleeding | Step 2

Most statins, including atorvastatin, are metabolized by CYP3A4, so CYP3A4 inhibitors such as clarithromycin and azoles raise levels and cause myopathy and rhabdomyolysis.

6 A Migraine without aura

Migraine | Step 1

Migraine without aura needs at least 5 attacks of 4–72 hours with 2 of unilateral, pulsating, moderate to severe, and activity-worsened pain, plus nausea or photophobia with phonophobia.

WHY NOT THE OTHERS

(C) Cluster attacks last 15–180 minutes with autonomic signs.

(E) Tension-type headache is bilateral, pressing, and not worsened by activity, without nausea.

7 (C) PSP, Alzheimer disease, or other pathologies can cause it

Atypical parkinsonism · Step 1

CBS is not specific, since PSP, AD, and other pathologies can cause it, and only autopsy confirms the protein.

WHY NOT THE OTHERS

(A) DAT SPECT is reduced in all three and does not identify the protein.

(B) CBD is a 4R tauopathy.

(E) The hot cross bun sign suggests MSA.

8 (A) Once IOP is below 40 mmHg

Glaucoma · Step 2

Acute angle closure is treated with acetazolamide, timolol, and apraclonidine, then IV mannitol if needed, and pilocarpine once IOP is below 40 mmHg; laser peripheral iridotomy follows.

WHY NOT THE OTHERS

(C) At very high pressure the ischemic iris sphincter does not respond.

(D) Pilocarpine constricts the pupil and helps open the angle.

9 (B) Aspirin plus clopidogrel for 21 days, then one antiplatelet

Physiology · Clots and bleeding · Step 1

After a minor stroke or high-risk TIA, aspirin plus clopidogrel is started within 24 hours for 21 days, then one antiplatelet continues, provided there was no thrombolysis and no cardioembolic source.

10 (C) It increases, inhibiting the thalamus and causing bradykinesia

Physiology · Basal ganglia pathways · Step 1

Loss of SNc dopamine weakens the direct pathway and strengthens the indirect pathway, increasing GPi inhibition of the thalamus and causing bradykinesia.

11 (D) Cluster headache

Cluster headache · Step 1

Cluster headache causes excruciating unilateral orbital pain for 15–180 minutes, often at night at the same time, with ipsilateral tearing, red eye, rhinorrhea, ptosis, and miosis, and makes patients restless.

12 (A) Rivastigmine

Dementia with Lewy bodies · Step 2

Acetylcholinesterase inhibitors, rivastigmine, donepezil, and galantamine, offset cholinergic loss and improve cognition, function, and hallucinations in DLB.

13 (E) Glycine

Physiology · Synapses and neurotransmitters · Step 1

GABA is the main inhibitory transmitter in the brain, and glycine is inhibitory mainly in the cord.

WHY NOT THE OTHERS

(A) Glutamate is the main excitatory transmitter.

14 (E) Mannitol or hypertonic saline

Herniation syndromes · Step 2

Raising the head to 30°, osmotic therapy with mannitol or hypertonic saline, and brief hyperventilation bridge to treating the cause, such as evacuation or decompressive craniectomy; LP is avoided with mass effect.

15 (A) Saccule

Vertigo · Step 1

The utricle and saccule sense linear acceleration, gravity, and head tilt, the utricle horizontal and the saccule vertical motion; the canals sense rotation.

WHY NOT THE OTHERS

(B) The horizontal canal senses yaw.

(C) The utricle senses horizontal linear acceleration.

(E) The cochlea transduces sound.

16 (B) Idarucizumab

Intracerebral hemorrhage · Step 2

Dabigatran is reversed with idarucizumab; apixaban or rivaroxaban with andexanet alfa or 4-factor PCC, heparin with protamine, and tPA with cryoprecipitate.

WHY NOT THE OTHERS

(A) Protamine reverses heparin.

(C) Vitamin K reverses warfarin.

(D) Cryoprecipitate reverses tPA.

(E) Andexanet reverses apixaban and rivaroxaban.

17 **A** **Greater petrosal nerve to the lacrimal gland**

Physiology · V and VII · Step 2

At the geniculate ganglion, the greater petrosal nerve leaves for the lacrimal gland; the chorda tympani carries taste and the stapedius branch prevents hyperacusis.

WHY NOT THE OTHERS

- C Only motor fibers leave the stylomastoid foramen.
- D Loss of the stapedius causes hyperacusis.
- E The chorda tympani carries taste and salivary fibers.

18 **B** **GPi inhibition falls, causing chorea**

Physiology · Basal ganglia pathways · Step 1

In Huntington disease, indirect-pathway neurons die first, reducing GPi inhibition and causing chorea.

WHY NOT THE OTHERS

- A This describes Parkinson disease.
- E Dopamine loss is the mechanism in Parkinson disease.

19 **E** **It raises plasma osmolality, drawing water out of the brain**

Physiology · Blood pressure and brain pressure · Step 1

Osmotic therapy with mannitol or hypertonic saline raises plasma osmolality, drawing water out of the brain across the BBB and lowering ICP within minutes. Mannitol is an osmotic diuretic that causes hypovolemia, acute kidney injury, and electrolyte shifts.

20 **B** **To prevent amblyopia**

Cataract · Step 2

Congenital cataracts follow genetic causes, intrauterine infection, and metabolic disease such as galactosemia, present with leukocoria, and are treated early because they cause amblyopia.

21 **E** **CRVO**

Vision loss and red eye · Step 1

CRVO, mostly after age 50 with hypertension, glaucoma, diabetes, or hyperlipidemia, gives a blood and thunder fundus of dilated, tortuous veins and hemorrhages in all quadrants.

WHY NOT THE OTHERS

- A Detachment brings flashes, floaters, and a curtain.
- C CRAO shows a pale retina with a cherry-red spot.

22 **D** **Left V1**

Physiology · V and VII · Step 1

The corneal reflex runs V1 in and VII out. A left V1 lesion stops the afferent signal, so neither eye blinks; a CN VII lesion would stop the blink on that side only.

23 **C** **DBS**

Typical Parkinson disease · Step 2

DBS treats refractory motor fluctuations.

WHY NOT THE OTHERS

- A Haloperidol blocks D2 receptors and worsens parkinsonism.
- B Tetrabenazine depletes dopamine.
- D Stopping levodopa would worsen parkinsonism.

24 **B** **Glucose and non-contrast CT head**

Acute ischemic stroke · Step 2

Glucose and non-contrast CT head come first, to rule out hypoglycemia and hemorrhage before thrombolysis. If there is no hemorrhage, CT angiogram looks for large vessel occlusion.

WHY NOT THE OTHERS

- C Aspirin is given within 24–48 hours, after hemorrhage is excluded.
- D For thrombolysis, BP needs only to be below 185/110.
- E Hemorrhage must be excluded before thrombolysis.

25 **C** **Chorea-acanthocytosis**

Huntington disease · Step 2

Chorea-acanthocytosis, autosomal recessive, causes orofacial chorea worse with eating, tongue biting, myopathy, and onset around 30, with acanthocytes on the blood film.

WHY NOT THE OTHERS

- B His HTT repeat length is normal.
- D Spinocerebellar ataxias cause prominent ataxia.
- E Wilson disease causes orofacial dystonia with parkinsonism.

26 **D** **Cerebral amyloid angiopathy**

Intracerebral hemorrhage · Step 1

Cerebral amyloid angiopathy is a common cause of lobar bleeds in older adults, may present with dementia, and shows lobar microbleeds on MRI with SWI. Hypertension bleeds in deep structures such as the basal ganglia, thalamus, pons, and cerebellum.

WHY NOT THE OTHERS

(B) AVMs are a cause in young patients.

(C) Hypertensive bleeds are deep, and she has no hypertension.

27 (A) Peripheral vertigo, with MRI if still unclear

Ruling out mimics · Step 2

Isolated vertigo with a reassuring HINTS examination suggests peripheral vertigo, with MRI if unclear.

WHY NOT THE OTHERS

(D) A reassuring HINTS examination points away from stroke.

(E) Hemorrhage would usually cause other signs or show on CT.

28 (B) Prompt high-dose glucocorticoids

Localizing a lesion · Step 2

A new headache over 50, with jaw claudication, vision loss, or raised ESR and CRP, suggests GCA, which needs prompt high-dose glucocorticoids to prevent blindness.

WHY NOT THE OTHERS

(A) Oxygen treats cluster headache.

(C) This is the workup for a thunderclap headache.

(D) Stopping analgesics treats medication overuse headache.

29 (D) Preventing delayed ischemia from vasospasm

Physiology · Blood pressure and brain pressure · Step 1

Nimodipine prevents delayed ischemic deficits, improving outcome more than vasospasm on imaging; its main adverse effect is hypotension. Nicardipine and labetalol control BP, and an antifibrinolytic is used only if repair is delayed.

30 (B) Multiple sclerosis

NMOSD · Step 1

Short, peripheral cord lesions, Dawson fingers, OCB, and a progressive course favor MS; OCB are usually absent in NMOSD and their presence is a red flag.

WHY NOT THE OTHERS

(A) MOGAD favors disc edema, conus lesions, and fluffy lesions.

(E) NMOSD usually lacks OCB and has long central cord lesions.

31 (E) Presynaptic voltage-gated calcium channels

Physiology · Synapses and neurotransmitters · Step 2

Lambert-Eaton syndrome targets presynaptic voltage-gated calcium channels, reducing ACh release, and causes proximal weakness, autonomic dysfunction, and reduced reflexes that improve with repeated contraction, often with small cell lung cancer.

WHY NOT THE OTHERS

(D) Myasthenia gravis targets postsynaptic AChR and causes fatigable weakness.

32 (E) Dorsal columns

Motor signs and syndromes · Step 1

Sensory ataxia follows dorsal column or large-fiber loss, with a stomping gait and a positive Romberg test, as in B12 deficiency. Cerebellar ataxia persists with the eyes open.

33 (B) JC virus antibody status

Physiology · MS, NMOSD, and MOGAD · Step 1

Natalizumab blocks α4 integrin, stopping lymphocytes crossing the BBB, but can cause PML from JC virus, so JC virus status is checked first.

34 (E) An early sign of ischemia

Choosing imaging and EEG · Step 1

Early ischemia under 12 hours often looks normal or shows loss of gray-white differentiation, insular ribbon blurring, or lentiform nucleus effacement.

WHY NOT THE OTHERS

(B) An old infarct is dark, like CSF.

(D) SAH shows bright blood in the subarachnoid space.

35 (A) It dilated the pupil, letting the bunched iris narrow the angle

Glaucoma · Step 1

Sympathomimetics such as adrenaline, phenylephrine, and pseudoephedrine dilate the pupil, and the bunched iris base can close the angle in predisposed eyes, such as hyperopic eyes with a shallow anterior chamber.

36 (D) Amitriptyline

Tension-type headache · Step 2

Amitriptyline prevents chronic tension-type headache, and frequent analgesic use risks medication overuse headache.

WHY NOT THE OTHERS

(B) Verapamil prevents cluster headache.

37 **(C) Decreased Aβ42 with increased tau and phosphorylated tau**

Alzheimer disease · Step 2

CSF in Alzheimer disease shows decreased Aβ42 with increased tau and phosphorylated tau, and amyloid PET shows plaques in life.

38 **(C) Duloxetine**

Physiology · Pain pathways and descending modulation · Step 2

SNRIs and TCAs, such as duloxetine and amitriptyline, block serotonin and norepinephrine reuptake in the spinal cord, strengthening descending inhibition and relieving neuropathic pain.

39 **(D) Oral prednisone 1 mg/kg for 7 days, then tapered**

Hearing loss · Step 2

Sudden SNHL, a loss of over 30 dB at 3 consecutive frequencies within 72 hours, is an emergency treated with oral prednisone, because spontaneous recovery is most likely in the first 2 weeks. MRI of the internal auditory canals is also obtained.

40 **(D) IV methylprednisolone for 3–7 days**

Multiple sclerosis · Step 2

An acute relapse is treated with IV methylprednisolone for 3–7 days, with plasma exchange if refractory.

WHY NOT THE OTHERS

(A) Plasma exchange is for relapses refractory to steroids.

(E) Baclofen treats spasticity.

41 **(E) Vasospasm and delayed cerebral ischemia**

Stroke symptoms and complications · Step 2

Subarachnoid hemorrhage is complicated by rebleeding, highest early before the aneurysm is secured, and by vasospasm and delayed cerebral ischemia on days 3–14. New focal deficits on day 6 without new blood fit vasospasm.

WHY NOT THE OTHERS

(A) Hemorrhagic transformation follows ischemic stroke and would show blood within an infarct.

(D) Her aneurysm is secured, and CT shows no new blood.

42 **(C) Left anterior temporal lobe**

Frontotemporal dementia · Step 1

The semantic variant follows left anterior temporal atrophy and causes fluent speech that has lost meaning, with anomia and poor single-word comprehension. The nonfluent variant follows left inferior frontal and insular atrophy.

WHY NOT THE OTHERS

(A) Right anterior temporal atrophy causes prosopagnosia and getting lost.

(B) Left inferior frontal atrophy causes effortful, agrammatic speech.

43 **(A) Temporal lobe**

Seizures · Step 1

Focal autonomic and psychological seizures arise in the temporal lobe and insula, with auras such as déjà vu and nausea. Involvement of the reticular formation or thalamus brings LOC, automatisms such as chewing and lip smacking, and a post-ictal state.

WHY NOT THE OTHERS

(B) Frontal lobe seizures are motor.

(E) An aura shows a focal onset; absence seizures have no post-ictal state.

44 **(C) Treat the cause and avoid fluid restriction**

Subarachnoid hemorrhage · Step 2

Hyponatremia after SAH follows SIADH or cerebral salt wasting. It is treated by treating the cause and avoiding fluid restriction, since vasospasm is treated with euvolemia.

WHY NOT THE OTHERS

(B) Fluid restriction risks hypovolemia during the vasospasm window.

45 **(C) Central cord syndrome**

Stroke symptoms and complications · Step 1

Central cord syndrome, from hyperextension injury and syringomyelia, causes arm weakness greater than leg, with the hands worst, cape-like pain and temperature loss, and sacral sparing.

WHY NOT THE OTHERS

(B) Posterior cord syndrome causes sensory ataxia with preserved strength.

(D) Anterior cord syndrome weakens all limbs below the lesion and spares vibration.

(E) Hemisection gives one-sided weakness with opposite pain loss.

46 **(C) Rituximab**

Physiology · MS, NMOSD, and MOGAD · Step 2

NMOSD is prevented by B-cell depletion with rituximab, IL-6 receptor blockers, eculizumab, or azathioprine. Interferon beta, natalizumab, and other MS drugs can worsen it.

47 **(A) It increases uveoscleral outflow**

Physiology · The eye · Step 1

Prostaglandin analogs such as latanoprost increase uveoscleral outflow and are first line for open-angle glaucoma. They darken the iris and lengthen eyelashes.

48 **(E) Uhthoff phenomenon**

Multiple sclerosis · Step 1

In Uhthoff phenomenon, symptoms worsen with heat. It is managed by avoiding heat and is not a relapse, which develops over days to weeks.

WHY NOT THE OTHERS

(A) Lhermitte sign is an electric shock down the spine on neck flexion.

(B) A relapse develops over days to weeks and does not settle within an hour.

49 **(D) Right hemisphere**

Localizing a lesion · Step 1

Eyes deviated toward one side suggest a hemispheric lesion on that side, as in MCA stroke.

WHY NOT THE OTHERS

(A) The eyes look toward a hemispheric lesion, here the right.

(E) Dorsal midbrain lesions impair upgaze.

50 **(E) Emergency craniotomy and evacuation**

Epidural hematoma · Step 2

Large or symptomatic epidural hematomas, or those with midline shift, need emergency craniotomy and evacuation; only small, stable ones with a high GCS are observed.

WHY NOT THE OTHERS

(D) LP with mass effect risks tonsillar herniation.

Test B answers

Answer key | Explanations

1 A	11 B	21 E	31 C	41 E
2 B	12 B	22 E	32 B	42 A
3 C	13 A	23 E	33 D	43 B
4 D	14 C	24 A	34 C	44 A
5 B	15 B	25 A	35 E	45 C
6 E	16 C	26 E	36 D	46 B
7 C	17 C	27 D	37 B	47 D
8 C	18 D	28 E	38 E	48 D
9 D	19 A	29 C	39 D	49 A
10 B	20 D	30 A	40 E	50 A

1 A It blocks VMAT2, depleting vesicular dopamine

Physiology | Movement disorders | Step 2

VMAT2 inhibitors such as tetrabenazine block loading of dopamine into vesicles, lowering chorea, but cause depression and suicidality, so mood is monitored.

2 B Ventilation-perfusion mismatch and the Haldane effect

Physiology | Respiratory center | Step 2

Oxygen raises CO_2 in COPD through ventilation-perfusion mismatch, as hypoxic vasoconstriction is reversed, and the Haldane effect, rather than loss of hypoxic drive, so saturations of 88–92% are targeted.

3 C Burr hole drainage

Subdural hematoma | Step 2

Symptomatic chronic subdurals are drained through burr holes, large acute ones are evacuated by craniotomy, and small ones without mass effect are observed.

4 D Clarke nucleus, inferior cerebellar peduncle

Physiology | Ascending tracts | Step 1

The dorsal spinocerebellar tract arises in the Clarke nucleus (C8–L3) and ascends uncrossed through the inferior cerebellar peduncle; the ventral tract arises from spinal border cells and enters through the superior peduncle.

WHY NOT THE OTHERS

(B) This describes the ventral spinocerebellar tract.

5 B Pneumonia

Alzheimer disease | Step 1

Complications include aspiration pneumonia from dysphagia and infection, with pneumonia the most common cause of death.

WHY NOT THE OTHERS

(E) Amyloid angiopathy can cause lobar ICH, but it is not the most common cause of death.

6 E Laser peripheral iridotomy

Glaucoma | Step 2

Acute angle closure is lowered with acetazolamide, timolol, and apraclonidine, then IV mannitol if needed and pilocarpine once IOP is below 40 mmHg. Laser peripheral iridotomy, which lets aqueous humor bypass the pupil, is the treatment of choice.

WHY NOT THE OTHERS

(C) Atropine dilates the pupil and can trigger or worsen angle closure.

(D) Laser trabeculoplasty treats open-angle glaucoma.

7 **C** **Dandy-Walker malformation**

Physiology · Nervous system embryology · Step 1

The Dandy-Walker malformation combines failure of the vermis to form with cystic enlargement of the fourth ventricle and a large posterior fossa; most patients develop hydrocephalus.

WHY NOT THE OTHERS

- (A) Holoprosencephaly is a forebrain malformation.
- (D) Chiari 2 is descent of the vermis and brainstem with myelomeningocele.
- (E) Chiari 1 is tonsillar descent with a small posterior fossa.

8 **C** **Uncal herniation compressing CN III**

Herniation syndromes · Step 1

In uncal herniation, the medial temporal lobe passes over the tentorium and compresses CN III, causing an ipsilateral blown pupil, ptosis, and a down-and-out eye.

9 **D** **It reaches cortex without a thalamic relay**

Physiology · I, II, and VIII · Step 1

Olfactory receptor neurons, which regenerate for life, pass through the cribriform plate to mitral cells in the olfactory bulb, which project to piriform cortex, amygdala, and entorhinal cortex without a thalamic relay.

10 **B** **Relative preservation of the medial temporal lobe**

Dementia with Lewy bodies · Step 1

In DLB, the medial temporal lobe is relatively preserved, so memory is spared early, while cortical involvement causes attention, executive, and visuospatial decline.

11 **B** **A venous thrombus crossing a right-to-left shunt**

Acute ischemic stroke · Step 1

Paradoxical embolus occurs with ASD, PFO, and atrial septal aneurysm, where a right-to-left shunt lets a venous thrombus reach the arterial circulation. With no other cause and age under 60, PFO closure is considered.

12 **B** **Propranolol**

Motor signs and syndromes · Step 2

Postural tremor of both hands at 6–12 Hz, often familial, is essential tremor, treated with propranolol or primidone.

13 **A** **Indirect-pathway medium spiny neurons projecting to the GPe**

Huntington disease · Step 1

Indirect-pathway neurons, which contain enkephalin and project to the GPe, are lost first, so the GPi inhibits the thalamus less and the released thalamus overdrives the cortex, causing chorea. Direct-pathway loss later causes rigidity and bradykinesia.

14 **C** **A swallow screen**

Acute ischemic stroke · Step 1

Supportive care comprises a swallow screen before oral intake, treating fever and glucose, and DVT prophylaxis, because dysphagia causes aspiration pneumonia.

WHY NOT THE OTHERS

- (B) A bubble echocardiogram looks for PFO.
- (E) Repeat imaging is for clinical worsening.

15 **B** **Corneal scraping, then urgent topical antibiotics**

Vision loss and red eye · Step 2

Bacterial keratitis is a sight-threatening corneal infection, most often after contact lens wear, from organisms such as Staphylococcus aureus and Pseudomonas. Corneal scraping for Gram stain and culture precedes urgent topical antibiotics such as moxifloxacin.

WHY NOT THE OTHERS

- (D) Supportive care is for viral conjunctivitis, which spares vision.

16 **C** **4**

Intracerebral hemorrhage · Step 2

The ICH score gives 1 point for GCS 5–12 (2 for 3–4), 1 for age 80 or older, 1 for infratentorial origin, 1 for volume of 30 mL or more, and 1 for IVH. He scores 1 + 1 + 0 + 1 + 1 = 4. The score estimates 30-day mortality.

17 **C** **Bilateral, high-frequency, downsloping SNHL**

Hearing loss · Step 1

Presbycusis causes difficulty in noise and loss of high-pitched consonants, with a bilateral, high-frequency, downsloping curve.

WHY NOT THE OTHERS

(A) A 4 kHz notch is noise-induced.
(B) An air-bone gap marks conductive loss.
(D) Low-frequency loss suggests Ménière disease.

18 (D) GJB2

Hearing loss · Step 1

Nonsyndromic loss is autosomal recessive in 75–80%, so parents are usually unaffected carriers. DFNB1 mutations in GJB2 and GJB6, encoding connexins 26 and 30, are the most common genetic cause of severe to profound congenital deafness.

WHY NOT THE OTHERS

(B) PAX3 causes Waardenburg syndrome types 1 and 3, with pigment changes.
(E) Merlin mutations cause NF2, with bilateral vestibular schwannomas.

19 (A) Left thalamus

Stroke symptoms and complications · Step 2

Central post-stroke pain often follows a thalamic infarct, though any sensory pathway lesion can cause it.

WHY NOT THE OTHERS

(B) A right cerebellar lesion would cause ataxia on the right, not pain.
(D) An occipital infarct causes visual field loss.
(E) The caudate is not a sensory relay.

20 (D) It delays central compensation

Physiology · Vertigo and the ear · Step 1

Vestibular suppressants such as meclizine block H1 and muscarinic receptors in the vestibular nuclei and relieve acute symptoms, but beyond a few days they delay central compensation, so vestibular rehabilitation follows.

WHY NOT THE OTHERS

(A) Intratympanic gentamicin destroys vestibular hair cells.

21 (E) Pain and temperature in the left leg

Physiology · Ascending tracts · Step 1

In Brown-Séquard syndrome, the dorsal columns and corticospinal tract are lost on the same side, and spinothalamic fibers, which cross within 1–2 segments of entry, are lost on the opposite side.

22 (E) Secondary progressive

Multiple sclerosis · Step 1

Gradual worsening after 10–15 years, with or without relapses, marks the secondary progressive course; primary progressive worsens from onset.

WHY NOT THE OTHERS

(A) CIS is a first episode.
(D) Primary progressive MS worsens from onset, without an earlier relapsing phase.

23 (E) Blood blocks CSF absorption at the arachnoid granulations

Subarachnoid hemorrhage · Step 1

Blood blocks CSF absorption at the arachnoid granulations, causing hydrocephalus, which is treated with an EVD.

24 (A) Subarachnoid hemorrhage from a saccular aneurysm

Subarachnoid hemorrhage · Step 1

PKD is one of the inherited conditions, with Ehlers-Danlos syndrome, fibromuscular dysplasia, coarctation of the aorta, and Marfan syndrome, linked with aneurysms, and these conditions need screening.

25 (A) CN IX

Physiology · IX, X, XI, and XII · Step 1

CN IX carries pressure and oxygen signals from the carotid sinus and body that slow the heart, and CN X carries parasympathetic fibers to the heart.

WHY NOT THE OTHERS

(C) CN X forms the efferent limb to the heart.

26 (E) Immunoglobulin levels before treatment and yearly

NMOSD · Step 2

With RTX, immunoglobulin levels are checked before treatment and yearly, because low levels raise infection risk.

WHY NOT THE OTHERS

(B) Titer does not predict severity or set the dose.

27 (D) Vascular dementia

Ruling out mimics · Step 1

Stepwise decline with vascular risk factors suggests vascular dementia, shown on MRI.

WHY NOT THE OTHERS

(A) Alzheimer disease declines gradually, with early memory loss.

(C) Prion disease progresses over months with myoclonus.

28 **(E) MOG-IgG is found in only 0.4% of MS, so false positives result**

MOGAD · Step 1

Because MOG-IgG is found in only 0.4% of MS, testing everyone with possible MS yields false positives.

WHY NOT THE OTHERS

(A) A live cell-based assay uses serum.

(B) MS drugs appear ineffective in MOGAD.

(C) It is rare in MS.

29 **(C) The dura is less firmly attached to the skull**

Epidural hematoma · Step 1

In young patients the dura is less firmly attached to the skull, so arterial blood strips it away more easily; in older adults atrophy stretches bridging veins, favoring subdural hematoma.

30 **(A) Non-contrast CT of the head**

Migraine · Step 2

A sudden, severe thunderclap headache is a red flag for SAH and needs non-contrast CT, then LP if negative, even in someone with migraine.

31 **(C) Right optic tract**

Physiology · I, II, and VIII · Step 1

Each optic tract carries the opposite visual field to the LGN, so tract and V1 lesions cause homonymous hemianopia.

WHY NOT THE OTHERS

(B) A Meyer loop lesion causes an upper quadrantanopia.

(D) A left tract lesion causes a right hemianopia.

(E) Chiasmal lesions cause bitemporal hemianopia.

32 **(B) Delirium**

Ruling out mimics · Step 1

Sudden onset, fluctuating attention, and an acute illness indicate delirium, which needs a search for its cause.

WHY NOT THE OTHERS

(A) Lewy body fluctuations occur over a chronic course with hallucinations and parkinsonism.

(C) Depression causes low mood and poor effort.

(D) Alzheimer disease develops over years.

33 **(D) Sorbitol accumulates and draws water into the lens**

Cataract · Step 1

Diabetes causes snowflake cortical opacities, because sorbitol accumulates and draws water into the lens.

WHY NOT THE OTHERS

(A) Urochrome collects in nuclear cataract.

(C) This is phacolytic glaucoma.

(E) Fibrous metaplasia occurs in subcapsular cataract.

34 **(C) Cataract in the right eye**

Cataract · Step 1

Cataract causes monocular diplopia as light refracts through clear gaps in the lens.

WHY NOT THE OTHERS

(A) Optic neuritis causes blurred vision with pain, not double images.

(B) Nerve palsies cause binocular double vision.

(E) Myasthenia causes binocular double vision.

35 **(E) It slows Aβ clearance**

Alzheimer disease · Step 1

Decreased Aβ clearance is represented by APOE ε4, which slows clearance and raises the risk of late-onset disease; increased production comes from APP, PSEN1, and PSEN2 mutations and Down syndrome.

WHY NOT THE OTHERS

(A) The presenilins form the catalytic part of γ-secretase.

(B) Extra APP gene dosage occurs in Down syndrome.

36 **(D) Simple febrile seizure**

Seizures · Step 2

Febrile seizures occur at 6 months to 5 years with fever of 38 °C or more and no CNS infection. Simple febrile seizures are a single generalized seizure of 15 minutes or less; complex ones have focal features, last 15 minutes or more, or recur within 24 hours.

37 **(B) Right jugular foramen**

Localizing a lesion · Step 2

CN IX, X, and XI together point to the jugular foramen. The right CN XI turns the head to the left and shrugs the right shoulder.

WHY NOT THE OTHERS

(C) The cerebellopontine angle involves CN V, VII, and VIII.

(E) The cavernous sinus holds CN III, IV, V1, V2, and VI.

38 (E) It causes supine hypertension

Physiology | Spasticity, bladder, and autonomic | Step 1

Midodrine, a prodrug of an α1 agonist, constricts arterioles and veins to raise standing BP and causes supine hypertension, piloerection, scalp itching, and urinary retention, so it is given during the day only.

39 (D) 100% oxygen by face mask

Cluster headache | Step 2

Cluster attacks are treated with 100% oxygen or subcutaneous or intranasal sumatriptan; oral triptans act too slowly for attacks under an hour. Verapamil is for prevention.

WHY NOT THE OTHERS

(A) Oral triptans act too slowly.

(B) Verapamil prevents attacks; it does not stop one.

40 (E) It blocks peripheral DDC, so more levodopa reaches the brain

Physiology | Movement disorders | Step 1

Carbidopa blocks peripheral DDC, so less levodopa is converted to dopamine outside the brain, lowering nausea and leaving more to cross the BBB.

WHY NOT THE OTHERS

(D) That describes entacapone, a COMT inhibitor.

41 (E) CN IX afferent, CN X efferent

Physiology | IX, X, XI, and XII | Step 1

CN IX carries touch from the back of the tongue and pharynx, forming the afferent limb, and CN X, from the nucleus ambiguus, supplies the pharynx, forming the efferent limb.

42 (A) Baclofen

Physiology | Spasticity, bladder, and autonomic | Step 2

Baclofen activates GABA-B receptors in the spinal cord and treats spasticity in MS, and its abrupt withdrawal causes seizures, hallucinations, and hyperthermia. It is renally cleared and causes sedation, weakness, and dizziness.

WHY NOT THE OTHERS

(B) Stopping fingolimod causes rebound disease, not seizures and fever within days.

43 (B) Inhibition of inspiration to prevent overinflation

Physiology | Respiratory center | Step 1

Slowly adapting stretch receptors mediate the Hering-Breuer reflex, in which inflation inhibits inspiration. Irritant receptors cause cough, and juxtacapillary receptors respond to edema.

44 (A) Pain is not worsened by routine activity

Tension-type headache | Step 1

Tension-type headache is bilateral, pressing, and not worsened by activity, without nausea, whereas migraine is unilateral, pulsating, worse with activity, and comes with nausea.

45 (C) No proven benefit, and behavior may worsen

Frontotemporal dementia | Step 2

Cholinesterase inhibitors and memantine have no proven benefit in FTD, and donepezil can worsen behavior; SSRIs can improve behavior.

WHY NOT THE OTHERS

(B) No treatment slows FTD.

(D) Benefit has not been shown.

46 (B) Increased perfusion at the focus

Choosing imaging and EEG | Step 2

Ictal SPECT shows increased perfusion at a seizure focus when injected within 90 seconds of onset, whereas FDG-PET shows low glucose use there between seizures.

WHY NOT THE OTHERS

(C) Striatal uptake is measured by DAT SPECT.

(D) Amyloid PET is for Alzheimer disease.

(E) Low uptake at the focus is the interictal FDG-PET finding.

47 (D) Review the antiseizure drug and start folic acid before conception

Physiology | Nervous system embryology | Step 2

The neural tube closes by day 28, often before pregnancy is recognized, so folic acid is started before conception. Valproate raises the risk of neural tube defects about tenfold and is avoided where possible in those who could become pregnant.

WHY NOT THE OTHERS

- (A) The neural tube has closed by 8 weeks.
- (B) Stopping all drugs risks seizures, and the tube has already closed by the time pregnancy is confirmed.
- (C) Folate, not B12, prevents most neural tube defects.
- (E) Screening detects defects but does not prevent them.

48 D Entacapone, a COMT inhibitor

Typical Parkinson disease · Step 1

Entacapone is a COMT inhibitor that prolongs each levodopa dose; rasagiline is a selective MAO-B inhibitor, and pramipexole is a D3-selective dopamine agonist.

WHY NOT THE OTHERS

- (A) Rasagiline inhibits MAO-B.
- (B) Trihexyphenidyl is an anticholinergic.
- (C) Amantadine is a weak NMDA antagonist.
- (E) Pramipexole is a dopamine agonist.

49 A PSP-P

Atypical parkinsonism · Step 2

PSP-P may respond to levodopa and, like PSP-PAGF, allows survival of a decade or more, whereas PSP-RS leads to death about 7 years from onset.

WHY NOT THE OTHERS

- (B) PSP-CBS is a cortical variant.
- (E) PSP-RS is the classic form with early falls and gaze palsy.

50 A Stop the suppressants and start vestibular rehabilitation

Vertigo · Step 2

Vestibular neuritis is treated with antiemetics, antihistamines such as meclizine, or benzodiazepines for about 3 days, then vestibular rehabilitation. Suppressants beyond a few days delay central compensation, and steroids lack evidence.

Test C answers

Answer key | Explanations

1 A	11 A	21 E	31 D	41 C
2 A	12 C	22 C	32 D	42 B
3 B	13 A	23 B	33 C	43 E
4 A	14 E	24 C	34 E	44 B
5 B	15 A	25 A	35 D	45 B
6 D	16 E	26 B	36 C	46 A
7 B	17 B	27 A	37 E	47 E
8 E	18 B	28 D	38 E	48 D
9 E	19 D	29 A	39 C	49 C
10 D	20 C	30 C	40 D	50 D

1 A Multiple system atrophy

Atypical parkinsonism | Step 1

MSA is an α-synucleinopathy with GCIs in oligodendrocytes. In PD and DLB, α-synuclein collects in neurons, and PSP and CBD are 4R tauopathies.

2 A Bridging veins

Subdural hematoma | Step 1

Subdural hematoma comes from bridging veins between the dura and arachnoid; atrophy stretches them so that minor trauma tears them.

3 B Left CN VI nucleus

Physiology | III, IV, and VI | Step 2

The CN VI nucleus is the horizontal gaze center, linked through the MLF to the opposite CN III nucleus, so its lesion abolishes gaze to that side in both eyes.

WHY NOT THE OTHERS

- (A) An MLF lesion causes INO with only adduction failure.
- (C) A nerve lesion weakens only the left lateral rectus.

4 A Right anterior temporal lobe

Frontotemporal dementia | Step 1

Right, nondominant, anterior temporal atrophy causes prosopagnosia, getting lost, behavior change, and memory loss.

WHY NOT THE OTHERS

- (B) Left inferior frontal atrophy causes the nonfluent variant.
- (D) Left anterior temporal atrophy causes the semantic variant.

5 B Pseudounipolar

Physiology | Neurons and action potentials | Step 1

Most neurons, including motor neurons, are multipolar; some are bipolar; and sensory neurons of the dorsal root ganglia are pseudounipolar.

WHY NOT THE OTHERS

- (A) Motor neurons are multipolar.
- (E) Bipolar neurons have two separate processes.

6 D Meningococcal vaccination

NMOSD | Step 2

Eculizumab blocks complement C5 and is given after meningococcal vaccination, since complement defends against Neisseria.

WHY NOT THE OTHERS

- (A) HLA-B*1502 testing is for carbamazepine.
- (E) JC virus testing is for natalizumab.

7 B ARIA

Physiology | Dementia and mood | Step 2

Anti-amyloid antibodies clear plaques but cause ARIA, edema or microhemorrhage, more often in

APOE4 carriers, so MRI monitoring is needed and anticoagulation is avoided.

8 **E** **It blocks NMDA channels noncompetitively**

Physiology · Dementia and mood · Step 1

Memantine blocks NMDA channels noncompetitively, limiting excitotoxicity, with modest benefit in moderate to severe disease. It is renally cleared.

9 **E** **NMOSD**

NMOSD · Step 1

CSF OCB are usually absent in NMOSD, and their presence is a red flag. Over 50 white cells/μL, or neutrophils or eosinophils, favor NMOSD over MS.

10 **D** **Phenytoin**

Seizures · Step 1

Phenytoin causes gingival hyperplasia, nystagmus and ataxia at high levels, SJS, and fetal hydantoin syndrome, and follows zero-order kinetics at high doses.

WHY NOT THE OTHERS

- (C) Valproate causes hepatotoxicity and neural tube defects.
- (E) Carbamazepine causes agranulocytosis and hyponatremia.

11 **A** **Concentric visual field loss**

Physiology · Antiseizure drugs · Step 2

Vigabatrin blocks GABA transaminase irreversibly and causes permanent concentric visual field loss, so vision is monitored. ACTH is the alternative for West syndrome.

12 **C** **Medullary reticulospinal tract**

Physiology · Descending tracts · Step 1

The medullary reticulospinal tract, mainly uncrossed, inhibits extensors, facilitates flexors, and contributes to autonomic and breathing control.

WHY NOT THE OTHERS

- (B) The pontine reticulospinal tract excites extensors.
- (E) The lateral vestibulospinal tract excites extensors.

13 **A** **Labyrinthitis**

Vertigo · Step 1

Vestibular neuritis spares hearing, whereas labyrinthitis adds unilateral hearing loss and tinnitus.

WHY NOT THE OTHERS

- (B) BPPV attacks last under a minute.
- (C) Vestibular neuritis spares hearing.
- (D) Ménière attacks last 20 minutes to 12 hours.

14 **E** **Posterior limb**

Physiology · Descending tracts · Step 1

The lateral corticospinal tract descends through the posterior limb of the internal capsule, whereas the corticobulbar tract passes through the genu.

WHY NOT THE OTHERS

- (C) The genu carries the corticobulbar tract.

15 **A** **Metoclopramide**

Physiology · When drugs cause disease · Step 2

Drug-induced parkinsonism follows striatal D2 block by antipsychotics and by metoclopramide and prochlorperazine, or dopamine depletion by tetrabenazine and reserpine, with symmetric, widespread signs and a normal DAT SPECT.

16 **E** **Mood and suicidal thoughts**

Huntington disease · Step 2

Tetrabenazine inhibits VMAT2, reducing dopamine release and lessening chorea. It can cause depression, suicidality, fatigue, and akathisia, so mood and suicidal thoughts are monitored.

17 **B** **Charcot-Bouchard aneurysms in penetrating arterioles**

Intracerebral hemorrhage · Step 1

Chronic hypertension causes lipohyalinosis and Charcot-Bouchard aneurysms in penetrating arterioles, which bleed in the basal ganglia, thalamus, pons, and cerebellum.

WHY NOT THE OTHERS

- (A) Saccular aneurysms cause subarachnoid hemorrhage.
- (E) Amyloid angiopathy causes lobar bleeds.

18 **B** **A seizure spreading along the motor cortex**

Localizing a lesion · Step 1

Symptoms spreading to the next body part are a Jacksonian march, a seizure spreading along the cortex.

WHY NOT THE OTHERS

- A A visual aura starts in the vision.
- C A cord lesion spares the face.
- D Ischemia causes sudden loss of function rather than a spreading positive symptom.

19 **D** **Observation, expecting it to resolve**

Seizures · Step 2

Todd paralysis is temporary weakness after a focal or generalized tonic-clonic seizure, lasting minutes to 1–2 days and resolving without lasting damage. Mistaking it for a stroke can lead to unneeded thrombolysis.

WHY NOT THE OTHERS

- E A witnessed seizure with tongue biting points to Todd paralysis, a stroke mimic.

20 **C** **Biconvex collection that stops at the sutures**

Epidural hematoma · Step 1

An epidural hematoma is a hyperdense, biconvex, lens-shaped collection that stops at the sutures, where the dura attaches, and can cross the midline.

21 **E** **Yes, the conus lesion is a supporting feature**

MOGAD · Step 2

A low positive titer of 1:10 to below 1:100 requires negative AQP4-IgG and at least one supporting feature, such as a conus lesion, with better diagnoses excluded.

WHY NOT THE OTHERS

- A Myelitis is one of the most common syndromes.
- B Low titers are accepted with a supporting feature.
- C Better diagnoses, including MS, must always be excluded.

22 **C** **Lower renal sympathetic tone and ANP release, causing natriuresis**

Physiology · Cardiovascular center · Step 1

Cardiopulmonary receptors sense volume: stretch lowers renal sympathetic tone and releases ANP, causing natriuresis, while low volume raises sympathetic tone, RAAS activity, and vasopressin.

WHY NOT THE OTHERS

- A Stretch lowers renal sympathetic tone.
- E This follows low volume.

23 **B** **Reduced sympathetic supply to the eye**

Cluster headache · Step 1

Cluster headache causes ipsilateral Horner signs, ptosis and miosis, from reduced sympathetic supply, alongside parasympathetic tearing and nasal congestion from the trigeminal autonomic reflex.

WHY NOT THE OTHERS

- C CN III compression causes a dilated pupil, as in uncal herniation.

24 **C** **ECG and orthostatic BP**

Ruling out mimics · Step 2

A trigger, lightheadedness, brief jerks, and rapid recovery suggest syncope, assessed with ECG and orthostatic BP.

25 **A** **Delirium**

Alzheimer disease · Step 2

Delirium is favored by sudden onset, fluctuating attention, and an acute illness, and prompts a search for the cause.

26 **B** **Every fiber crosses before leaving the back of the midbrain**

Physiology · III, IV, and VI · Step 1

Every CN IV fiber crosses before leaving the back of the midbrain, so a nuclear lesion weakens the opposite superior oblique.

WHY NOT THE OTHERS

- A CN VI runs free in the cavernous sinus.
- D CN III leaves through the interpeduncular fossa.
- E The CN VI nucleus is the horizontal gaze center.

27 **A** **Tension-type headache**

Tension-type headache · Step 1

Tension-type headache, the most common primary headache, is bilateral, band-like, pressing, mild to moderate, not worsened by activity, and free of nausea, often with pericranial muscle tenderness.

28 **D** **Less salt, caffeine, and alcohol, with a thiazide or betahistine**

Vertigo · Step 2

Ménière disease, from endolymphatic hydrops, causes attacks of 20 minutes to 12 hours with fluctuating low-frequency loss, tinnitus, and fullness. It is treated with less salt, caffeine, and alcohol, thiazides or betahistine, intratympanic steroids, and intratympanic gentamicin if refractory.

29 **A** **JC virus status**

Multiple sclerosis · Step 2

Natalizumab can cause PML, so JC virus status is checked first.

WHY NOT THE OTHERS

(E) Bradycardia on starting is a concern with fingolimod, through S1P receptors on the SA and AV nodes.

30 **C** **Morphine shifts it right; propofol flattens it**

Physiology · Pain and anesthesia · Step 2

Opioids shift the CO_2 response curve right, whereas benzodiazepines and propofol flatten it, and together they add to respiratory depression.

31 **D** **3 sodium out and 2 potassium in**

Physiology · Neurons and action potentials · Step 1

The Na/K-ATPase pumps 3 sodium out and 2 potassium in to maintain the gradients.

32 **D** **Ventral pontine (Millard-Gubler) syndrome**

Stroke symptoms and complications · Step 1

Ventral pontine (Millard-Gubler) syndrome causes contralateral hemiplegia, ipsilateral CN VI and CN VII palsy with the forehead included, and horizontal diplopia.

WHY NOT THE OTHERS

(A) AICA syndrome spares strength and adds hearing loss.

(B) Foville syndrome causes a horizontal gaze palsy rather than an isolated abduction deficit.

(C) Wallenberg syndrome causes no hemiparesis.

(E) Weber syndrome causes a CN III palsy.

33 **C** **Increases uveoscleral outflow**

Glaucoma · Step 1

Prostaglandin analogs such as latanoprost increase uveoscleral outflow and are first line; they lengthen eyelashes and darken the iris.

WHY NOT THE OTHERS

(B) This is how carbonic anhydrase inhibitors work.

(D) This is how pilocarpine works.

(E) This is how timolol works.

34 **E** **Systemic lidocaine toxicity, then arrhythmia and cardiac arrest**

Physiology · Pain and anesthesia · Step 2

Local anesthetics block voltage-gated sodium channels; systemic toxicity starts with perioral numbness and seizures and progresses to arrhythmia and cardiac arrest. Beta-blockers and cimetidine raise lidocaine levels, and amide allergy is rare.

35 **D** **T2 shine-through, not acute infarct**

Choosing imaging and EEG · Step 2

Old edema or gliosis can be bright on DWI, T2 shine-through, but true acute infarct is dark on ADC.

WHY NOT THE OTHERS

(A) Acute infarct is dark on ADC.

(E) Enhancement is seen on T1 with gadolinium.

36 **C** **Topical antibiotic drops such as chloramphenicol**

Vision loss and red eye · Step 1

Bacterial conjunctivitis causes mucopurulent discharge with lids stuck together, often bilaterally, with normal vision; it is often self-limited, and topical antibiotics shorten it.

WHY NOT THE OTHERS

(B) Antihistamines treat allergic conjunctivitis.

(D) Steroids treat anterior uveitis.

37 **E** **Ventilation tubes**

Hearing loss · Step 2

Otitis media with effusion is the most common cause of acquired hearing loss in children, and ventilation tubes treat persistent effusion.

WHY NOT THE OTHERS

(A) This is an acquired conductive loss.

(B) Stapes surgery treats otosclerosis.

(C) Prednisone is for sudden SNHL.

(D) Implants are for severe bilateral SNHL.

38 E Medication overuse headache

Migraine · Step 1

Frequent acute treatment causes medication overuse headache, so acute drugs are limited to fewer than 10–15 days a month and prevention is offered.

39 C Check and relieve bladder distension

Physiology · Cardiovascular center · Step 2

Autonomic dysreflexia follows SCI at or above T6. A stimulus below the lesion, usually a full bladder (85%), drives splanchnic vasoconstriction, while vagal bradycardia and flushing occur only above the lesion. Fixing the catheter or bladder is often life-saving.

40 D Low-dose quetiapine

Dementia with Lewy bodies · Step 2

Antipsychotics are avoided whenever possible, because serious reactions occur in up to 50%; if one is essential, low-dose quetiapine is used, and haloperidol, risperidone, olanzapine, and aripiprazole are generally avoided.

WHY NOT THE OTHERS

- (A) Haloperidol is generally avoided.
- (B) Aripiprazole is generally avoided.
- (C) Olanzapine is generally avoided.
- (E) Risperidone is generally avoided.

41 C Right subthalamic nucleus

Motor signs and syndromes · Step 1

Hemichorea and hemiballismus classically follow a lesion of the contralateral subthalamic nucleus, most often from ischemic stroke over 65, and tend to resolve.

42 B The opposite cerebral peduncle is pressed against the tentorium

Herniation syndromes · Step 1

Kernohan notch: the opposite cerebral peduncle is pressed against the tentorial edge, causing hemiparesis on the same side as the lesion, a false localizing sign.

43 E Lens proteins leak and macrophages clog the trabecular meshwork

Cataract · Step 2

In phacolytic glaucoma, a hypermature lens leaks proteins, and macrophages clog the trabecular meshwork while the angle stays open; it is treated by removing the lens.

WHY NOT THE OTHERS

- (A) This is phacomorphic glaucoma, with a closed angle.
- (B) Pupillary block closes the angle.

44 B C8–T1 injury from an apical lung tumor

Motor signs and syndromes · Step 2

Klumpke palsy, a C8–T1 injury from hyperabduction traction or an apical lung tumor, causes intrinsic hand weakness and clawing with medial forearm sensory loss, and Horner syndrome if sympathetic fibers are involved.

WHY NOT THE OTHERS

- (C) An ulnar lesion does not cause Horner syndrome or forearm numbness.
- (D) A medullary stroke does not cause hand wasting.
- (E) C5–C6 injury causes Erb palsy.

45 B T-type calcium channels

Physiology · Antiseizure drugs · Step 1

Ethosuximide blocks T-type calcium currents in thalamic neurons, which drive thalamocortical spike-wave rhythms, and is first line for absence seizures. Valproate weakly blocks the same channels.

46 A Allow it up to 220/120

Acute ischemic stroke · Step 2

Without thrombolysis, BP is allowed up to 220/120 in ischemic stroke, whereas it is kept below 185/110 for thrombolysis and actively lowered in intracerebral hemorrhage.

WHY NOT THE OTHERS

- (C) A systolic target of 140 is for intracerebral hemorrhage.
- (E) Nimodipine prevents delayed ischemia after subarachnoid hemorrhage.

47 E Systolic below 160 with IV nicardipine or labetalol

Subarachnoid hemorrhage · Step 2

BP is controlled with IV nicardipine or labetalol until the aneurysm is secured, keeping systolic below 160.

WHY NOT THE OTHERS

- (A) That threshold is for ischemic stroke without reperfusion.
- (B) That threshold applies before thrombolysis.

(D) This is the long-term target after hypertensive ICH.

48 **(D) FLAIR**

Choosing imaging and EEG · Step 1

FLAIR makes lesions bright with fluid made dark, showing periventricular plaques, SAH in the sulci, and infarcts after 4.5–6 hours.

WHY NOT THE OTHERS

(A) T1 makes fat and gadolinium bright.
(C) MRA shows flowing blood.
(E) SWI shows blood products as dark.

49 **(C) Add entacapone**

Typical Parkinson disease · Step 2

As neurons are lost, 'on' time shortens, and entacapone, a COMT inhibitor that prolongs each levodopa dose, or extended-release levodopa is added. DBS treats refractory motor fluctuations.

50 **(D) It blocks peripheral conversion of levodopa to dopamine**

Typical Parkinson disease · Step 1

Levodopa crosses the BBB and becomes dopamine, and carbidopa blocks its peripheral conversion; together they are the most effective treatment.

WHY NOT THE OTHERS

(C) Rasagiline is the MAO-B inhibitor.
(E) Entacapone is the COMT inhibitor.